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KENYATTA UNIVERSITY

INSTITUTE OF OPEN LEARNING

SZL 100: GENERAL ZOOLOGY

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Objectives of the course

1. To introduce the students to the natural philosophy of science (and more of biology}.
2. To equip the students with the knowledge of modern scientific methodological procedure.
3. To equip the students with the background knowledge to enable them fit all into any other specialized area of zoology.

REFERENCES.



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LESSON 1

ORIGIN OF THE UNIVERSE

1.1 Introduction

The current space explorations seem to suggest that life exist in other planets. The sun, the earth, and the other planets belong to the solar system that occurs towards the outer edge of the Milky Way. The Milky Way is just one of the over 10 billion galaxies. A galaxy is any of the large star systems held together by gravitational pull. A Milky Way is a defuse band of light stretching across our sky with millions of faint stars and is visible on clear nights.

1.2 Objectives

At the end of this lesson, you should be able to:

- ☐ State the theories of origin of the universe
- ☐ Describe the origin of the earth
- ☐ Discuss the theories of origin of life

1.3 The origin of the Earth

One theory that tries to explain the origin of the earth is the 'Big Bang' theory. According to this theory, formation of the earth the earth is thought to have began about 20 billion years ago. At about this time, the cosmic expansion was initiated and a mighty explosion is thought to have occurred. All matter that was seen present began to spread apart at a great rate. The early universe was very hot, very dense and perhaps also very irregular. Within minutes, after the Big Bang some nuclear reactions occurred. The noise of the original explosion can still be 'heard' in the form of background radiation permeating all of space, a trace that appears as 'snow' on a television receiver that is tuned to a channel that is not receiving a picture.

As the matter in the universe cooled, it eventually condensed into galaxies, according to one scenario for the evolution of the universe. The galaxies fragmented into stars and clustered together to form great aggregations over vast regions of space. As the first generations of stars



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were born and died, the heavy elements, such as carbon, oxygen, silicon, and iron were gradually synthesized. As stars evolved into real giants, they ejected matter that condensed into dust grains. New stars formed from clouds of gas and dust. In at least one such nebula, the cold dust collapsed into a thin disk surrounding the star. Dust grains adhered to one another by coalescence and accumulated into larger bodies that grew in size by their gravitational attraction, forming the diverse array of bodies, from tiny asteroids to giant planets that constitute the solar system.

1.4 The Origin of Life

The present earth is thought to have formed about 4.5 billion years ago from swirling cosmic dust and gases. The oldest rocks containing traces of living things are dated about 3.5 billion years ago. These rocks, called 'Warrwoona Group' is found in the South Pole in Australia. They contain microscopic fossils of spherical prokaryote cells filamentous bacteria and structures very much like formations being produced today by cyanobacteria.

Various hypotheses have been put forward to explain how life first began on earth. The major theories include:

1. Special creation: this theory proposes that life was created by a supernatural being at a particular time.
2. Spontaneous generation: this theory proposes that life arose from non-living matter on numerous occasions.
3. Steady- state theory: this theory states that life has no origin.
4. Cosmozoan theory panspermia theory: this proposes that life arrived on this planet from else where.
5. Biochemical evolution: this proposes that life arose according to chemical and physical laws.

1.4.1 Special creation

This theory is upheld by most of the world's major religions and civilizations and attributes the origin of life to a supernatural event at a particular time in the past. The traditional Judeo-Christian account of creation given in genesis 1: 1-26 has attracted and continues to attract



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controversy. All Christians agree that the bible is Gods word to man. Unlike science, which broadly relies on observation and experiment to seek the truth, theology draws its insights from divine revelation and faith.

Faith, according to Hebrews 11: 1-3, is the substance of things hoped for, the evidence of things not seen.....by faithwe understand that, the universe was created by Gods word, so that what can be seen was made out of what cannot be seen. Faith accepts things for which there is no evidence in the scientific sense. This means that logically there can be no conflict between scientific and theological accounts of creation since both are mutually exclusive.

Scientific truth to the scientists is a tentative but theological truth to the theist is absolute. Science concerns itself only with observable phenomenon and as such will never be able to prove or disapprove special creation.

1.4.2 Spontaneous generation

Aristotle (384 – 322 BC) believed that life arose spontaneously. He was the first Greek philosopher to use inductive method of reasoning in making his conclusions. He appealed directly to nature for his facts and wrote about 146 books. The most important of the books was the '**ladder of nature**' or '**scala naturae**' which is a taxonomic scheme whereby nature is viewed as a continuum from inanimate matter through plants to lowly animals to higher animals ending with mankind. Although this hierarchical system may have suggested an evolutionary history behind all living things, Aristotle still strongly subscribed to the prevailing view of the origin of biological 'form' through spontaneous generation that is i.e. rats and mice arose from old mice and man and other organisms arose from mud.

In 1688, Francesco Redi, an Italian biologist and physicist living in Florence questioned the theory of spontaneous generation. He observed that the little white worms seen on decaying flesh were fly larvae. By a series of experiments, he produced evidence to support the idea that life can arise only from pre- existing life the concept of **biogenesis**. He prepared eight flasks with a variety of kinds of meat inside. Four flasks were sealed and four left open to the air. Meat in the open vessel bred maggots while meat in sealed vessels decayed and decomposed but



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developed no maggots. It seemed that the maggots developed not out of meat but out of eggs of flies.

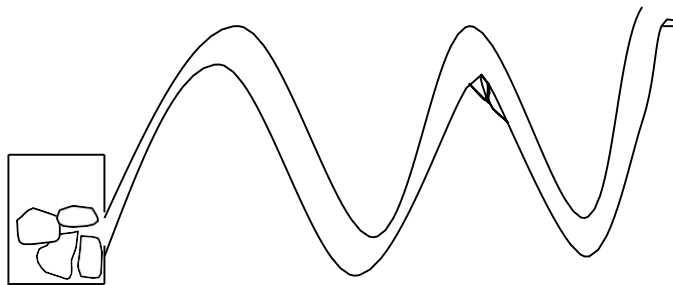
Activity 1. From the experiment of Redi described above, state the part of his experiment, which was the control and experimental unit. Replication is a key element in experimental research. How did Redi meet this criteria and why was it important?

Whilst Redi's experiments appeared to refute the spontaneous generation of flies, the pioneering work in microscopy by **Antonie Van Leeuwenhoek** appeared to reinforce the theory with regard to microorganisms. His observations with the microscope provided fuel for both biogenesis and spontaneous generation theories, finally stimulating other scientists to design experiments to settle the question of the origin of life by spontaneous generation.

In 1765, **Lazzaro Spallanzani**, who was an Italian biologist, prepared a nutritive solution, which he boiled for between one-half and three-quarters of an hour, then sealed the flasks. Microorganisms did not appear and he concluded from this that the high temperatures had destroyed all forms of living organisms in his vessels and without their presence no life could appear.

Activity 2. Suggest another reason why Spallanzani's experiment might have prevented the growth of organisms.

In 1860, Louis Pasteur (French, 1860) modified Spallanzani's experiment. He used a flask with a long S-shaped unsealed neck.





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Air could reach the broth but heavier particles of dust and bacteria are trapped in the flask neck. There was no growth of organisms and no decay of the broth.

1.4.3 Steady-State theory

This theory asserts that the earth had no origin, has always been able to support life has changed remarkably little if at all and that species had no origin. The theory proposes that species have always existed and that in the history of a species the only alternative is for its numbers to vary or for it to become extinct.

The theory does not accept the palaeontological evidence that presence or absence of a fossil indicates the origin or extinction of the species represented. It quotes as an example the case of the **coelanth**, *Latimeria sp.* Fossil evidence indicates that coelacanth died out at the end of the **Cretaceous** period, 70 million years ago. But a living specimen was discovered off the Coast of Madagascar, altering this view. The steady state theory claims it is only by studying living species and comparing them with the fossil records that extinction can be assumed and then there is high probability that this may be incorrect.

1.4.4 Cosmozoan theory

This theory does not offer a mechanism to account for the origin of life but favors the idea that it had an extraterrestrial origin. It does not therefore, constitutes a theory of origin as such, but merely transposes the problem to else where in the universe.

The theory states that life could have arisen once or several times and in various parts of the Galaxies of the universe. Its alternative name is the theory of **Panspermia**. Repeated sightings of UFOs, cave drawings of rocket-like objects and 'spacemen' and reports of encounters with aliens provide the background evidence for this theory.

1.4.5 Biochemical Evolution

Experiment by Francesco Redi (1668), Lazzaro Spallanzani (1729-1799) and Louis Pasteur (1862), demonstrated that life originated from other life. The question is then, if life only originated from other life, then where did the first life come from?



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Most scientists believe that the sun and the moon formed at the same time and in the same part of the solar system approximately 4.5 billion years ago. Life however, appeared much later about 3.5 billion years ago, when the first igneous rock was formed. Many biologists believe that the original state of the earth bore little resemblance to its present – day form and had the following probable appearance:

- (a) it was hot (about 4000-8000 °C) and as it cooled carbon and the less volatile metals condensed and formed the earth's core.
- (b) The surface was probably barren and rugged as volcanic activity, continuous earth movement and contraction and cooling folded and fractured the surface.

The atmosphere is believed to have been totally different in those days. It is believed that the spontaneous chemical reactions of hydrogen, water, ammonia and methane occurred in the atmosphere with lightening, radio active emissions, cosmic rays and UV acting as sources of energy. These elements combined in various ways to form various organic compounds including amino acids. From this there must have been the formation of organic soap which contained amino acids and energy rich compounds which are related to life.

The main factors, which may have favored accumulation of energy rich compounds, were:

- i) Absence of free oxygen ii)

Absence of living organisms

The atmosphere is believed to have been a **reducing atmosphere**, as indicated by the presence of the metals in their reduced form (such as iron (II) in the oldest rock of the earth. Laboratory experiments have also shown that it is far easier to generate organic molecules (the basis of living organisms) in a reducing atmosphere than an oxygen rich atmosphere.

In 1953, Stanley Miller, while doing his Ph.D. at the University of Chicago, in a series of experiments stimulated and proposed conditions of primitive earth. In his experimental high-energy chamber (Fig 1.1), he successfully synthesized many substances of considerable biological importance, including amino acids, adenine, and simple sugars such as ribose.



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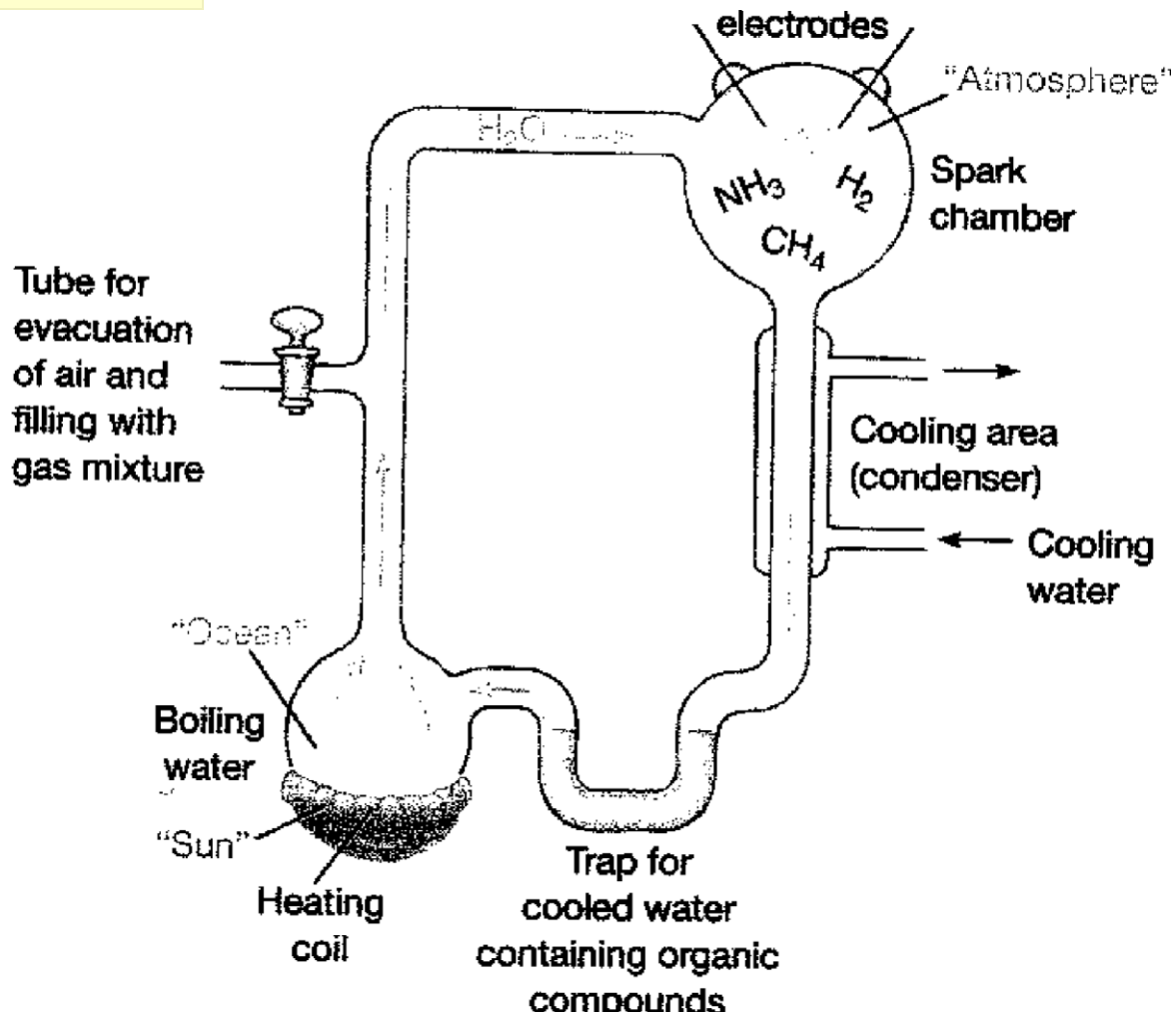


Fig 1.1. Stanley Miller's experiment in which he synthesized amino acids from gases under conditions thought to have been present in the primeval atmosphere. The gases and vapors were circulated under pressure and exposed for a high voltage for one week. At the end of the period the liquid products in the trap were analyzed by paper chromatography. A total of 15 amino acids were isolated including glycine, alanine and aspartic acid.

1.5 The nature of the earliest organisms

Following the formation of the organic soup, the stage was then set for the formation of the nucleic acid e.g. the DNA. The DNA occurring in the organic soup must have been naked, living



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in a sort of form of stored energy. The DNA must have the capacity to replicate and multiply, setting the stage for organic evolution. Current evidence suggest that the first organisms were the heterotrophs comprising bacteria which are thought to have given rise to blue-green algae which then initiated the **oxygen evolution**.

These algae, like true plants but unlike photosynthetic bacteria use water as the **electron source** in **non-cyclic photophosphorylation** and hence release molecular oxygen as a by- product. The evolution of bacteria to blue-green algae is known as the Precambrian evolution.

1.6 Summary of the theories of the origin of life

There have been two types of explanations for the origin of the universe.

- (a) Supernatural explanation- that either God or other supernatural bodies is responsible.
- (b) Naturalistic explanation – this includes the idea that organisms arose spontaneously from mud, called spontaneous generation. Experiments by Francesco Redi (1668), Lazzaro Spallanzani (1729-1799) and Louis Pasteur (1862) demonstrated from other life.
2. It is believed that life may have originated in primeval organic soup which resulted from spontaneous chemical reactions of hydrogen, water, ammonia and methane, utilizing energy from lightening, radio active emissions, cosmic rays and ultra violet (U.V) radiation.
3. The earliest life must have consisted of heterotrophic bacteria, which could exist, in the anoxic environment-giving rise to the blue-green algae, which release oxygen through biochemical processes. This resulted in the gradual build up of oxygen and its ionization to form the ozone layer would reduce the ultra violet radiation striking the earth. Whilst decreasing the rate of synthesis of new complex molecules, the decrease in radiation would confer some stability or otherwise successful forms of life.

Revision questions

1. What is our galaxy called? Milky way
2. How many galaxies are there? Over 10 billions.
3. When did the “Big Bang” occur? About 20 billions years ago.



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4. Name the theories put forward to explain how life first began on earth.
Special creation spontaneous generation, steady-state theory, Cosmozoan or Perspermia, Biochemical evolution.
5. For how long did Stanley Miller leave his experiment to run 7 days.
 - i. Explain how the Cosmic expansion (“Big Bang”) was initiated and what were the results.
 - ii. Explain briefly what the following terms mean:
 - Cyanobacteria
 - Biogenesis
 - Prokaryotes
 - Warrwoona Group
 - Eukaryotes
 - Reducing atmosphere
 - iii. name the factors that may have favored accumulation of energy rich compounds in the prebiotic atmosphere.
 - iv. Describe briefly using a large labeled diagram Millers Experiment to explain chemical evolution.
 - v. How are Cyanobacteria able to survive in aerobic conditions? (hint use: of “heterocysts”
6. Discuss the conditions that must be met in experimental research.

Further reading

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LESSON 2 THE CELL

2.1 Introduction

A cell is a living unit with which all animal (or plant) tissues are constituted. The cells are different in appearance in different parts of the body, some are oval, some are round, some are



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elongated, and some are columnar and so on. Except a few kinds like the stripped muscle cells, all other are so small that we need a microscope to see them.

If we imagine a cell with all the structures of different types of cells incorporated into one, we shall get a typical cell, which is described below.

2.2 Structure and functions of an animal cell

A typical animal cell is a minute blob of living matter. It is like a thin bag with viscous fluid inside. The bag is made of an extremely thin membrane- the cell membrane or plasmalemma. Within this bag is a viscous fluid- cytoplasm, In the middle of the cytoplasm there is a ball- like nucleus. It is a compact surrounded by a thin membrane- the nuclear membrane. Inside the nucleus there is a viscid fluid_ the nucleoplasm. In it there are some strands and granules called chromatin material.

Inside the nucleus there is usually a rounded body) the nucleolus. Within the cytoplasm of the cell, outside the nucleus, there are many organelles such as mitochondria, endoplasmic reticulum or ergastoplasm, microsomes, centrioles.

Golgi complex, food vacuoles and contractile vacuoles, all of which are seen as minute rods, granules or membranes with he light microscopes.

Only with an electron microscope we can see them in great detail. A description of the organelles as seen through an electron microscope is as follows: _

1. Cell-Membrane (Plasmalemma)

Although it is the outermost membrane of a cell, it is often described as an organelle. It is an extremely thin membrane, 75_105 A0 thick. It is a three layered structure having two layers two layers of protein molecules with an intervening lipid layer. The lipid molecules are arranged as bimolecular layer (lipid bilayer).

The hydrophilic ends of the lipid molecules, where the phosphate groups are located, touch the protein layers and the hydrophobic ends, where the fatty acids are located, touch one another in the middle of the cell membrane.



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2. Endoplasmic reticulum (ergastoplasm)

In its typical form it is a branching and anastomosing network of membranes forming tubes and sacs in the cytoplasm of a cell. The tubes

and sacs are filled with fluid. The space inside this membrane specially the wide ones are called cisternae. Sometimes the endoplasmic reticulum may appear as isolated fluid filled vesicles. The number of tubules, cisternae and vesicles may vary from cell to cell, and in some cell at different phases of physiological activity. The membranes of endoplasmic reticulum are of three types:

- a) **Granular or rough:** The surface of these membranes has minute globular bodies sticking to it called ribosome.
 - b) **Non-granular or smooth:** The surface of these membranes is smooth i.e. the ribosome is absent.
 - c) **Annulate lamellae:** These are stacks of membranous sacs with small pores on the membranes.
3. **Ribosome:** These are small dense granules of about 0.015 micrometers in diameter, lying freely in the cytoplasm or on the surface of the endoplasmic reticulum. These granules contain a significant proportion of ribonucleic acid (RNA) of the cell. When ribosome are in-groups, lying freely or attached to membranes, the groups are called polyribosomes.
4. **Mitochondria:** In the meshes of the endoplasmic reticulum, there is another kind of minute bodies called mitochondria. These are usually sausage-shaped, but some may be round or thread like. The biggest ones are 1.0 micrometer wide and 3.0 micrometers long. In pancreatic cells they are threads like and up to 10 micrometer long. A double membrane covers each mitochondrion; the inner being folded to form partitions projecting into the cavity of the mitochondrion. These partitions are called cristae. Inside the mitochondrion there are enzymes required for the citric acid cycle. Here, in the mitochondria, the hydrogen atoms are removed from the intermediate metabolites of citric acid cycle to unite them with oxygen to form water. In this process energy is released



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which is utilized in transforming adenosine diphosphate (ADP) and inorganic phosphate into adenosine triphosphate (ATP). This reaction, in which ATP is regenerated, is called oxidative phosphorylation. The ATP

then provides energy for various body functions.

5. Lysosomes

Scattered in the cytoplasm of a cell there are minute particles, called Lysosomes. These are packets of enzymes such as acid phosphatase, RNAase, DNAase and betaglucuronidase etc.

6. Microbodies (peroxisomes)

These are extremely minute bodies scattered in the cytoplasm of a cell. They contain the enzyme catalase. Microbodies are more frequently found in liver and kidney cells.

7 Cytoplasmic fibrils

These are fine threads like structures composed of protein molecules. They form a kind of a skeleton for the cell. These fibrils are specially seen near the intercellular connections

8. Microtubules

These are fine tubules with a diameter of 0.23 micrometers and of indefinite length.

These tubules probably form the spindle fibers when the cell divides.

9 Golgi Apparatus (Golgi Complex).

On one side of the nucleus there is an accumulation of cistern (membrane bound spaces) closely packed into stacks. This is called the Golgi Apparatus. In between the cisternae there are fine tubes and sacs or vesicles. The membranes of Golgi apparatus are without ribosome. Research has shown that the Golgi apparatus is concerned with the secretion of the cells.

10 Centrioles

In between the nuclear membrane and the Golgi apparatus there is clear patch of cytoplasm. This area was earlier called centrosome. In the middle of this area there are two dark bodies called Centrioles. Each centriole is a cylindrical structures 0.5



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micrometers long and 0.15 micrometers in diameter. In cross section each centriole is seen to be composed of nine sets of Microtubules arrange in circle.

Each set is again composed of three tubules. Before cell division, the Centrioles divide and occupy positions at opposite poles of the nucleus. Centrioles appear to play a role in the formation of spindle fibers when a cell divides.

11 Nucleus

It is usually a rounded body situated in the middle of the cell. In some cells of the liver, stomach and urinary bladder there may be more than one nucleus. A single striped muscle cell may have more than 100 nuclei. The nucleus is surrounded by a nuclear membrane (nuclear envelope) which is composed of two trilaminar membranes, inner and outer, separated by a gap of 0.5 micrometers.

This envelope is provided with pores (nuclear pores) which may have a diameter of .05 to .07 micrometers. At places the outer membrane is continuous with the endoplasmic reticulum. Inside the nuclear membrane is a fluid called nucleoplasm (nuclear sap) containing chromatin material which aggregates into denser masses immediately below the nuclear membrane called heterochromatin, and the material that remains scattered in the nucleoplasm is called euchromatin.

In the dividing cell the chromatin material forms thread like chromosomes. The chromatin is composed of DNA (deoxyribonucleic acid) in association with nucleohistones, acidic proteins and a variety of enzymes like RNA polymerase.

12 Nucleolus: The nucleus contains, besides the chromatin, one or two rounded bodies. These are composed of granules in the form of threads called nucleolonema and the ground substance in which these granules are placed is called pars amorpha. Major constituent of the nucleolus is RNA (ribonucleic acid). The nucleolus does not have a surrounding membrane.



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13 Microsomes

These are not independent organelles of the cytoplasm. Generally the broken parts of endoplasmic reticulum, ribosomes, membranes of Golgi complex and other membranes obtained by cell fractionation are together called microsomes or microsomal fraction.

The cytoplasm, in which the organelles are placed, is a viscous fluid consisting of water and various organic and inorganic molecules. The secretory granules and other particles, which are not permanent features of a cell, are not called organelles.

Activity

Draw and label the ultrastructure of an animal cell.

Revision questions.

- I. Describe using diagrams the lipid bilayer of cell membrane.
- II. Tabulate the cell organelles indicating their structure and function.
- III. Distinguish between: (a) Rough and smooth endoplasmic reticulum (b) Lysosomes and Microsomes.
- IV. Give a detailed description of mitochondrial structure and function.
- V. Cells of onion root tips have cells with more mitochondria than other parts. Explain.

2.3 Biological membranes

Introduction

The plasma membrane and other subcellular membranes are thin, pliable bilayers of phospholipids interspersed with proteins. One of the most important functions of a membrane is the regulation of movement of materials from one compartment to another. The substances move by simple diffusion, facilitated diffusion and active transport. In simple diffusion, the direction of flow is determined by the concentration gradient of each solute. In active transport, the membranes include pumps that use energy to push specific solute molecules across the membrane, against a concentration gradient. In facilitated diffusion, the solute molecules combine with a carrier molecule in the membrane.



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2.3.1 Structure and composition

The molecular structure of biological membranes consists of a phospholipid bilayer with proteins embedded in the bilayer and carbohydrates attached to proteins or phospholipids on the external surface of the membrane. This is the fluid mosaic model of membrane structure.

The lipids serve as a barrier to the passage of most materials between the inside and outside of the cell or organelle. They also account for much of the physical integrity of the membrane. Some membrane proteins allow selected materials to pass through the membrane while others receive chemical signals from the cell's external environment and respond by regulating certain intracellular processes. Some catalyze chemical reactions on the membrane surface. Others recognize specific molecules that impinge upon them. Carbohydrates are involved in recognition reactions.

Activity

Illustrate the fluid mosaic model of the double membrane

2.3.2 Activities of membranes

Biological membranes have a number of roles:

1. Compartmentalization of cells: Lipid bilayers form a physical barrier holding the cell contents in.
2. Regulation of traffic between compartments: The membrane is selectively permeable to materials. Both the lipid bilayer and some proteins effect this selective permeability.
3. Active pumping of solutes
4. Mediation of energy trapping and energy –releasing reactions. Some of the cell's metabolic enzymes are integrated into or attached onto the cell membrane.
5. Recognition and binding of materials at the cell surface. Receptors (glycoproteins) detect information coming in the form of drugs, toxins or hormones.
6. Membranes serve as a site for ribosome attachment.
7. Some proteins contribute to the cell's shape, mechanical strength and attachment to the neighbouring cell.



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8. In nerve cells, the plasma membrane is a conductor of nerve impulses from one end of the cell to another.
9. The membrane of muscle cells are electrically excitable.

LESSON 3

RESISTANCE TO DISEASES

3.1 Introduction

In this chapter, you will learn about the mechanisms used by animals to resist invasion by foreign microorganisms. The environment is alive with biological threats such as pathogenic (disease-producing) organisms. To cope with such threats, all living organisms are equipped with a number of specific or non-specific defensive mechanisms.

3.2 Objectives

By the end of this lesson, you should be able to:

- ☐ Differentiate between specific and non-specific defense mechanisms
- ☐ Discuss the non-specific defense mechanisms
- ☐ State the functions of the specific immune system
- ☐ State the difference between cellular and humoral immune responses

3.3 The Immune system

The immune system serves to defend the body from attacks by microorganisms. It also serves to maintain the integrity or self identity of the body by removing damaged cells or by clearing debris or foreign materials from the body's internal environment. Immunity is the body's capacity to resist all types of organisms or toxins that damage the tissues or organs.

The body's defense system is comprised of two components: specific and non-specific.

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3.3.1 Non-specific defense mechanisms (Innate immunity)

This component offers immediate protection from micro-organisms and other foreign particles. It does not discriminate between different invaders

(i.e. it acts non-specifically, protecting the animal against many different pathogens). Under these mechanisms, animals defend themselves against pathogens by preventing foreign organisms from entering the body and by destroying them if they enter. The following mechanisms apply:

The skin

The skin is the largest organ of the human body. It is the main protection against infection. The epidermis has a tough $\square$ keratin which cannot be easily digested by microorganisms. It also produces sweat which is acidic and inhibits the growth of pathogens. Sebum from the sebaceous glands is bacteriocidal while the commensal flora on the skin surface competes with the pathogens, inhibiting their growth. Besides, the skin is constantly renewed, repairing any damage and maintaining a constantly renewed barrier against infection.

Body secretions

Mucous membranes cover body surfaces that must be kept moist. They line the nasal cavity, digestive tract, urethra and vagina, secreting mucus that traps microorganisms. The microorganisms in the respiratory tract are trapped by cilia and swept by cilia into the pharynx. They are swallowed, enter the stomach and are encountered by mucus, acid and enzymes. The ears also produce wax (cerumen) that traps foreign particles while saliva and tears contain lysozyme that lyses bacterial cells.

Phagocytosis

Phagocytes are white blood cells that destroy pathogens and dead tissues by engulfing and digesting them. There are neutrophils, monocytes, and macrophages that travel throughout the body. Some macrophages are resident in the lungs and brain.



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Natural killer cells

A special population of white blood cells called natural killer cells destroy cancer cells and cells infected by viruses.

Inflammation

An inflammation is a local reaction to tissue damage or injury. It is characterized by swelling, redness, pain and heat. It helps to fight infection and heal the wound.

Question

How does an inflammation arise? How does it protect against foreign particles?

Proteins

Proteins that take part in the body's non-specific defense mechanisms include complement and interferons. **Complement** consists of some proteins produced by the liver that lyse bacteria, attract phagocytes to infection sites and enhance phagocytosis. **Interferons** are glycoproteins, produced by virus infected cells that interfere with replication of viruses and protect nearby cells from further attack. **Interleukins** are proteins that promote inflammation, stimulate release of neutrophils and cause an increase in amino acids in the blood, to be used in making other proteins that aid in eliminating pathogens.

3.3.2 Specific defense mechanisms (Acquired or adaptive immunity)

The primary function of the immune system is to discriminate between self and non-self. These mechanisms offer no immediate protection on first encounter with a pathogen. It acts on invaders that elude all the non-specific defense mechanisms. This arm of the immune system recognizes, selectively eliminates and remembers foreign invaders. It is specific. Adaptive immunity can be distinguished into two types:

- Humoral response, mediated by antibodies in the serum (blood plasma). It is used against bacteria and viruses that have not invaded cells. It is mediated by B-lymphocytes that produce antibodies.



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□ Cellular response, based on the production of cells that are sensitive to non-self. It acts against viruses, parasites and fungi that have invaded cells as well as foreign tissues. It is mediated by T-lymphocytes.

Note

Immunization (vaccination) is a form of acquired immunity. It induces the immune system to produce cells ready to fight a particular pathogen. It can be natural or artificial, passive or active.

Activity

Describe the following forms of immunity:

- (a) Passive natural immunity
- (b) Active natural immunity
- (c) Passive artificial immunity
- (d) Active artificial immunity



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LESSON 4

THE THEORY OF EVOLUTION AND THE FOSSIL RECORD

Welcome to Unit 5. In this unit, you will learn about the various evolutionary theories.

4.1 Introduction

During the primeval earth, the DNA occurring in the organic soup must have had the capacity to replicate and multiply, setting the stage for organic evolution. The environmental aspects playing a part at the level of the gene limited the multiplication level of that nucleic acid. The concept of competition then set in among these genes. Only those genes, which were fitting within that environment, could multiply. At the level, these were limited resources, which is the fundamental basis of organic evolution. In organic evolution, the stronger individuals survive and the weaker ones die out.

4.2 Objectives

At the end of this lesson, you should be able to:

- ☐ Outline the events in geological history in relation to the evolution of various animal groups
- ☐ Describe various methods of dating fossils
- ☐ Explain the causes of adaptive radiations and extinctions during geological history.

4.3 The geological time –scale

The history of the earth is divided for convenience into four geological eras (Archeozoic, Paleozoic, Mesozoic and Cenozoic) and 11 periods (Table 2.1). The two most recent periods (Tertiary and Quaternary) are further divided into seven systems or **epochs**.

The rocks of the earth's crust are stratified meaning lying layer upon layer. Unless disrupted by earth's movement, the earth gets progressively younger towards the top of a series of layers (strata). The sequence of fossil reveals a gradual increase in complexity of organisms



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from the lower strata to the highest, indicating that over geological periods of time some organisms have advanced in complexity.

Table 4.1. The Geological Time Scale

ERA	PERIOD	EPOCH	AGE (myrs)	Animal life	Plant life
Archeozoic	Precambrian		5000?	Origin of earth?	
			3500	Origin of life?	
			3000	Blue-green algae and bacteria (procaryotes)	
			2000	Primitive eukaryotes	
			1000	Primitive metazoans	
Paleozoic	Cambrian		570	Origin of all non-vertebrate phyla and echinoderms	
	Ordovician		500	Origin of vertebrates, jawless fish; trilobites; molluscs and crustaceans abundant	
	Silurian		440	Origin of jawed fish; earliest coral reefs	Earliest spore-bearing, vascular plants
	Devonian		400	Origin of Amphibia and ammonites; spiders appear; adaptive radiation of fish (cartilaginous and bony fishes)	Earliest mosses and ferns
	Carboniferous		350	Origin of reptiles and insects; adaptive radiation of amphibia	Abundance of tree-like ferns e.g. <i>Lepidodendron</i> , forming 'coral reefs'
	Permian		280	Adaptive radiation of reptiles; beetles appear; extinction of trilobites	Origin of conifers
Mesozoic	Triassic		225	Dinosaurs appear; adaptive radiation of reptiles	Abundance of cyads and conifers
	Jurassic		195	Dinosaurs dominant; Origin of birds and mammals; insects abundant	Origin of flowering plants
	Cretaceous		135	Extinction of ammonites and dinosaurs; origin of modern fish and placental mammals	Dominance of flowering plants
	Tertiary	Palaeocene	65	Horses, cattle and elephants	
		Eocene	54	Adaptive radiation of mammals	



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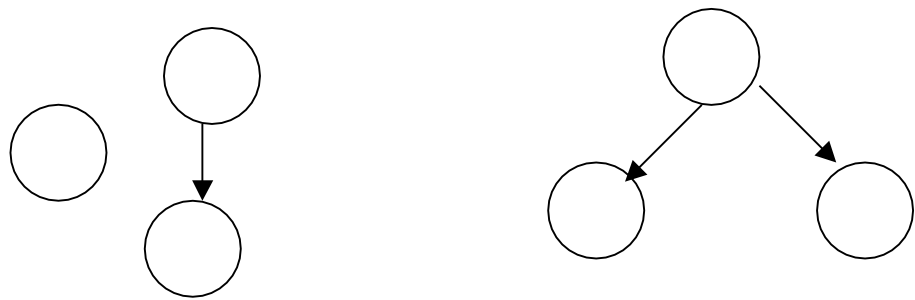
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Cenozoic		Oligocene	38	Dogs and bears appeared Apes and pigs appeared	Adaptive radiation of flowering plants, especially herbaceous types
		Miocene	26		
	Quaternary	Pliocene	7	Origin of humans	
		Glacial (Pleistocene)	2		
		Recent (Holocene)	0.01	Dominance of humans	

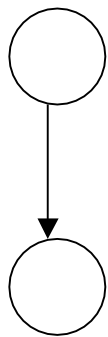
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Radioactive dating has established an approximate age for the oldest rocks belonging to each period. Three principal methods of radioactive dating are currently used:

- 1) Uranium/thorium methods: Uranium and thorium are generally found occurring together in the same rocks.



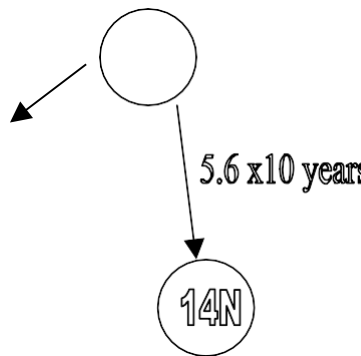
- 2) The potassium/argon method



- 3) Radiocarbon dating methods

4.4 Patterns in record.

About 600 million years ago, during the Cambrian period all phyla of marine invertebrates that entered the fossil record were of relatively complex organisms, with some phyla which are extinct today, being



particle

faunal and floral richness in the fossil



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represented. It is during this period, when the first land plants and insects appeared. Before the Cambrian period, the world was populated virtually by bacteria.

The period before the Cambrian is called the **Precambrian** period. During this period there, are very few fossils that have been recovered. There are several reasons that have been proposed as explanations of this paucity in fossils.

1. Most of the Precambrian organisms was soft-bodied and hence did not readily form fossils. It is the hard parts of plants and animals that are fossilized, because there are the most likely to resist decay and become buried.

This explanation though partly correct seems insufficient. It seems that many hard-bodied animals present at the Cambrian period arose suddenly at the time, from Precambrian ancestors that gradually developed shells or exoskeletons or other hard parts.

2. It is possible that the environmental conditions during the Pre-Cambrian period made fossilization unlikely. Fossils may be destroyed by normal geological processes, and the greater the lapse of time, the more likely the destruction. It is therefore likely that a high percentage of fossils formed as long as the Precambrian is likely to have been destroyed long before there were any human beings to study them.

4.5 Life explosions during the Cambrian

The Cambrian period marks the biggest burst of evolutionary history. The question is how such a sudden increase in living things occurred during the Cambrian period. Several hypothesis have been put forward to explain the life explosion, one of which is the **Berkner- Marshall Hypothesis**. This is one of the most provocative hypotheses. It was put forward by Berkner and Marshall of the South West Center for Advanced Studies, Dallas, Texas.

The two proposed that during the 1.7 byrs ago or so that photosynthetic organisms existed during the Pre- Cambrian, the concentration of molecular oxygen (O₂) in the atmosphere was still so low that living things could survive only at some depths in lakes or seas. This is because intense ultraviolet radiation at the surface or on land would have been lethal to living things. In



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In addition, the low concentration of oxygen would have imposed severe limits on anaerobic respiration.

According to Berkner-Marshall Hypothesis, the atmospheric oxygen concentration passed the “first critical level,” i.e. it reached about 1% of the present atmospheric levels at the beginning of the Cambrian, 600 myrs ago. This amount was enough to cut the intensity of ultra violet radiation so that life could exist anywhere in the earth’s bodies of water, and also enough to greatly enhance the evolutionary potential of aerobic respiration. The result was the explosive evolutionary surge that marked the beginning of the Cambrian period.

The hypothesis further stipulates that the great increase in living things during the Cambrian meant a far greater abundance of green plants, and hence much faster photosynthetic production of oxygen. The rapid rise in atmospheric concentration of oxygen that must have followed presumably reached a second “critical level” of about 10% of present levels some 420 million years ago, during the **Silurian period**. At this point, the spread of life into land became possible and thus a second great evolutionary explosion would have begun.

By the end of Paleozoic era, during the **Permian period (280myrs ago)**, there occurred a dramatic decline in the number of families of shallow- water invertebrates. It is hypothesized that the late Permian was a time when the earth’s continents coalesced to produce the single super – continent of **Pangaea**. The joining of continents produced a marked reaction in the area occupied by shallow seas (which occur round the periphery of continents) and thus a marked decline in the area of habitat available to shallow water invertebrates. After the Permian, there was an increase in faunal richness as the area occupied by shallow seas increased as the continents spread again.

Activity

- 1.Explain briefly the Berkner- Marshalls hypothesis.
- 2.Name the periods when the first and second critical levels in atmospheric oxygen were reached according to Berkner- Marshall hypothesis.



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4.6 Evolution of higher life forms

The modern theory of evolution states that organism have arisen by decent and modification of more ancient forms of life. For instance, it is believed that human beings evolved from now extinct animals resembling apes and that this happened through accumulation of changes from generations. In modern terms we can say that evolution is the process by which organisms of one generation come to differ from those of preceding generations.

There are many agents that may cause organisms to change from one generation to another, among which the most important is, **natural selection**. Other agents of evolution are mutations, sexual recombination, immigration, random drift or non-random mating.

4.7 Evolution by natural selection

Natural selection can be defined as differential reproductive success of individuals within a population that occurs because of genetic differences among them. The essence of natural selection is that some members of a population have more progeny that live to reproduce than others. If for example a physical barrier separates some individuals of a population, which have been interbreeding, the new population would find itself in a somewhat different environment. In so doing, some individuals will find it difficult to survive in the new environment and would die. Only the fit ones would survive and reproduce. The weak ones would be selected against by nature.

In the new environment, natural selection would favour other characteristics in the original population. In other words, the most adaptive characteristics in the original population may not be the most adaptive characteristics in the new environment. Eventually, as natural selection act on individuals in each of the two populations, the differences between them will become greater such that they can no longer interbreed, even if the geographical barrier between them should disappear. It is such a process, which after being repeated over billions of years has given rise to the radically different groups of organisms that inhabit the earth today.



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Darwin developed the concept of natural selection in 1859, using the following pieces of evidence and logic.

1. That all living things have enormous reproductive potential. For example if a single pair of house flies were to breed in April, and if all their offspring survived to reproduce, and their progeny offspring likewise survived, by August, there would roughly be 191 billion house flies. Human population also shows a clear illustration of geometric population growth.
2. The fact that populations tend to remain relatively stable demonstrates that most offspring die long before they reproduce.
3. This mortality is not shared equally. There are hereditary differences among individuals that attribute to their ability to survive and reproduce. Those individuals that carry genes, which produce damaging effects, are continually removed from populations, while those that posses superior characteristics survive.
4. Because of their improved probability of survival (based on the hereditary makeup), some organisms are able to breed more often and more successfully and produce a disproportionately large number of offspring. In otherwords, there will be a tendency of natural selection for more fit individuals within a population.

Activity 3. a) Explain how natural selection may act upon some individuals of the same population separated by a physical barrier.

b) Discuss briefly the argument used by Darwin in development of the idea of natural selection.

Summary

1. The history of he earth can be divided for convenience into 4 geological eras, which are further divided into eleven periods (Tertiary and Quaternary) are further subdivided into seven epochs.
2. There are very few fossils occurring during the Pre Cambrian period, but there appears to have been a life explosion during the start of the Cambrian period. This is a time, when according to Berkner-Marshall hypothesis, oxygen in the atmosphere reached the “first critical level”, i.e.1% of the present atmospheric levels.



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3. Evolutionary process occurs through a change of organisms from simple to more complex form through generations. The main agency of evolution is natural selection. Through natural selection, some individuals are able to

leave descendants than others. The concept

of natural selection was developed by Charles Darwin in 1859, when he published his small book “ On the origin of species by Natural Selection”

Revision questions

1. Discuss the development of the theory of evolution up to Darwin’s time.
2. Why are the following geologic times (periods) known as:
 1. Cambrian – Age of echinoderms
 2. Ordovician –age of molluscs.
 3. Silurian –age of coral reefs
 4. Devonian –age of fishes
 5. Carboniferous – age of amphibians
 6. Permian – age of birds



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LESSON 5

MICRO AND MACROEVOLUTION

Welcome to lesson 3. In this lesson you will learn about the concepts of micro- and macroevolution and the classification systems of animal species.

5.1. Introduction

The evolutionary concept dates back to 611 BC when some ancient Greeks and Roman Philosophers put forward the idea that all living things were related and that they had changed over time. Charles Darwin formalized the evolutionary concept as a theory, through his book “On the Origin of Species by Natural Selection” published in 1859. And although Darwin recognized that organisms change from primitive to more complex forms, he could not explain how evolution occurred. Darwin had no answer as to how variations were passed on from one generation to the next and this to wait until the early 19th Century when the Austrian monk, Gregor Mendel discovered that genes were the culprits. This marked the beginning of the field of genetics.

5.2. Objectives

At the end of this lesson, you should be able to:

1. Explain the meaning of micro- and macro-evolution.
2. Discuss the binomial system of zoological nomenclature.
3. Differentiate between phylogenetic and phenetic classification.
4. Apply dichotomous keys to animal species identification.

5.3. Micro-evolution

Evolution that occurs within species is called microevolution. It gives rise to geographic races and ecotypes, and is mediated by micromutations. Evolutionary changes in populations are ordinarily visualized as gradual, built upon many small genetic variations that arise and are passed on from generations to generation. The shifting gene frequencies in local populations may be thought of as microevolution. The shifting gene frequencies in a population can be



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described using the Hardy-Weinberg law This law is named after two people, G.H.Hardy. A British Mathematician and W.Weineberg, a German Geneticist.

The law states that the relative frequencies of the genes in a population will remain constant from one generation to the next if:

1. The population is large.
2. There is no selection for or against any specific gene or allele i.e. if mating occurs at random.
3. No mutations occur
4. There is no migration of individuals into or out of the population.

The operation of the Hardy-Weinberg principle will result in maintaining a given gene frequency in a population. The essential feature of the process of evolution is a gradual change in gene frequencies of a population when this Hardy-Weinberg equilibrium is upset either because:

1. Mutations occur
2. Reproduction are non-random.
3. The population is small, so that the gene frequencies in successive generations will be determined by chance events

5.4. Macro-evolution:

Macroevolution is a term, which refers to the evolutionary development of taxonomic units at the species level and above. The evolution of these units is the basis of the generation of taxonomic variety. Groups of organisms sharing a few major features in common can be subdivided into groups sharing few major features plus other features and these groups can be divided into more subgroups, and so on. This leads to the system of classification.

During the history of various taxa, periods of rapid evolution alternated with periods of stasis, when little change occurred. Mass extinctions occurred 6 times and were followed by extensive adaptive radiations. Extinctions were caused by:

- ☐ Movements of continents (Plate-tectonic theory), leading to changes in sea levels
- ☐ Changes in climate
- ☐ Volcanicity



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- ☐ Collision of asteroids

Adaptive radiation occurred due to:

- ☐ Natural selection as barriers were opened for dispersal
- ☐ Changes in the physical and biological environment
- ☐ Changes in the level of oxygen

5.5. Classification

Classification is the grouping of organisms on the basis of degrees of similarity between them. It is the process by which organisms may be identified and assigned to taxonomic groups or taxa (e.g. class). The science of classification is called **taxonomy**, also defined as the **theory** and practice of classifying organisms. It involves the naming of organisms (**nomenclature**) and the systematic placing of organisms into groups or taxa (systematics) on the basis of certain relationships between organisms. **Systematics** is the scientific study of the diversity of organisms, with the goal of assessing relationships among them.

Nomenclature is the practice of assigning names to organisms. In Zoology, the system of names used and the application of such names is governed by the International Code of Zoological Nomenclature. Biological nomenclature is based on the **binomial system** pioneered by the work of the Swedish naturalist Carl Linnaeus (1707_78). In 1735, Linnaeus published his book “Systema Naturae”, in which he used for the first time a binomial system of nomenclature. The binomial system of nomenclature postulates that the name of the animal or plant must consist of only two words (binomen); Example:

1. Mosquitoes, *Anopheles gambiae*
2. Man, *Homo sapiens*
3. Coffee plant, *Coffea arabica*

The first word designates the genus to which the animal or plant belongs. The second word denotes a particular *species* within that genus. It is known as the **specific epithet**. The term epithet refers to a “**meaningful term**”. The two words in combination comprise the species name and they form a **binomen**. The binomen is a **binary epithet**. Linnaeus is therefore considered the founder of **taxonomy**, the study of classification of species of living things.



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The binomial system classifies organisms into groups at various hierarchical levels on the basis of easily observable, shared structural features such as shape, number of, and position of limbs and so on.

5.5.1 Principles of classification

- (a) The name given to any organism must be in accordance with binomial nomenclature code.
- (b) The name must be in Latin, and if not, it must be Latinized.
- © The date of publication of the name must be given
- (d) Every animal is given a generic and a specific name, and sometimes a varietal name. (e) The name of a genus can be used only for one or a number of related species. It is started in upper case and italicized or underlined.
- (f) Specific epithets may be combined with different generic names and used for a number of unrelated organisms. It is always written in lower case and italicized or underlined.
- (g) A species given the first adequate description takes priority and the name used in that description is the valid name. This is the rule of priority.
- (h) When a species is first described, a holotype is deposited in a museum for later comparison. This is the **type species**.

5.5.2 Significance of biological classification

1. It is an aid in memory
2. It Improves our predictive powers
3. Provides relatively stable, unique and unequivocal names for organisms.
4. Provides a simple practical means by which students of Zoology may know what they are talking about.
5. Discovery of new knowledge: Classification results in new knowledge on the variation and distribution of these attributes.
6. Correct identification of species
7. For information storage, retrieval and communication
8. Helps in construction of phylogenetic relationships



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5.6. Ranks of taxa

A taxon is a taxonomic group or unit of any rank. Each animal or plant belongs to a series of taxa of consecutive ranks. The basic taxa are the following (proceeding from the most inclusive to the least inclusive):

Animal: Kingdom, Phylum, Class, Order, Family, Genus, Species

Each taxon may contain a number of taxonomic units below it, for example, one phylum may contain six classes or a genus three species but each taxon can belong to the single taxon immediately above it, for example an order can only belong to the class above it as shown below. An example of hierachial taxonomic groups based on the Linnean system of classification is given in figure 3.1.

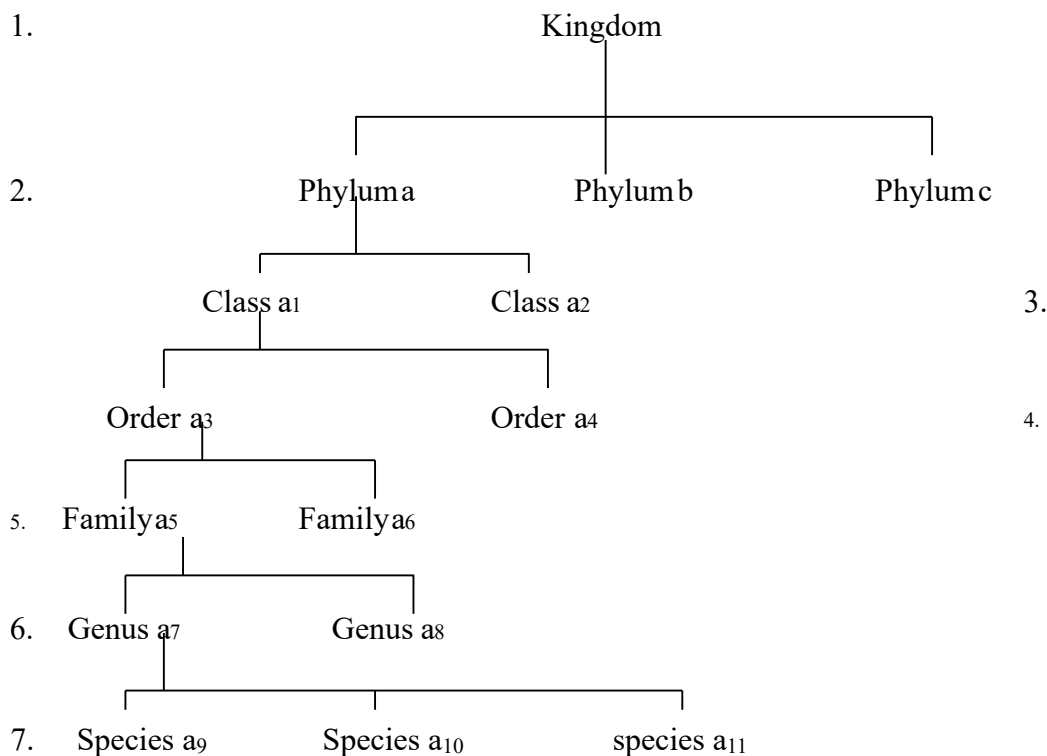


Fig. 3.1. A hierachial taxonomic group of classification.

Taxa may be further subdivided into subtaxa, such as subclass or grouped together into **supertaxa** such as **superclass**. Other taxa may also be formed by splitting taxa into **intrataxa** or **tribes** or grouping taxa into **cohorts**.



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Example: man

Kingdom	Animalia	
Subkingdom	Metazoa	
Phylum	Chordata	
Superclass	Gnathostomata	
Class	Mammalia	
Subclass	Theria	
Infraclass	Eutheria	
Order	Primates	
Suborder	Anthropoidea	
Superfamily	Hominoidea	
Family	Hominidae	
Subfamily	Homininae	
Genus :	Species:	Subspecies
<i>Homo</i>	<i>sapiens</i>	: <i>sapiens</i>

By convention, the names of some groupings are always given the same endings thus:

Super families: -oidea (Hominoidae)

Families: -idae (Hominidae)

Subfamilies: -inae (Homininae)

Tribes : -ini

Subtribes: -ina

Activity

Give a full classification of the following animals.

- Mosquitoe
- African elephant
- Locust
- Plasmodium
- Honey bee
- House fly



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5.7. Classification system

There are two types of classification systems: **artificial** and **natural**.

(i) **Artificial (or arbitrary classification):** This is based upon one or few easily observable characteristics, and is devised and used for a limited number of special purposes.

Example: Fish can be classified either as fresh water fish, marine fish or brackish-water fish, on the basis of the environment.

(ii) **Natural classification:** This may either be:

a) Phylogenetic or

b) Phenetic

(a) **Phylogenetic classification** reflects possible evolutionary relationship based on ancestry and descent. It is the most commonly used form of classification. In such a system, organisms belonging to the same taxa are believed to have a common ancestor, and may be represented in the form of a family tree called a **cladogram** (Fig 3.2.).

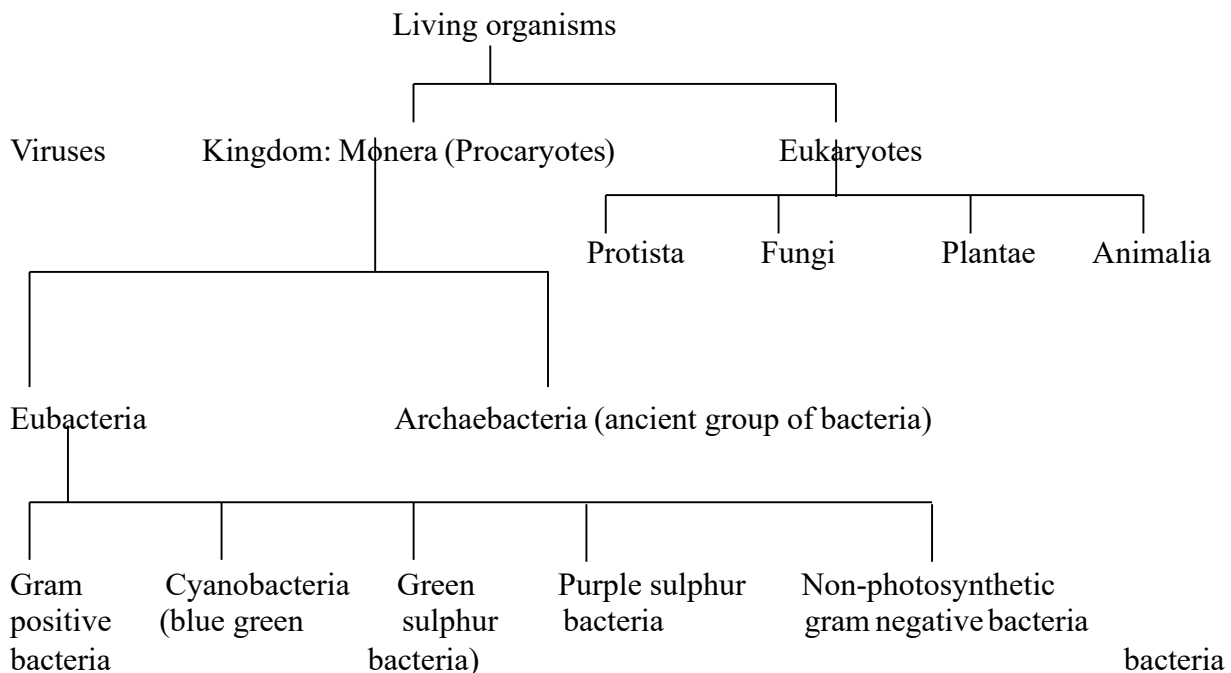


Fig. 3.2. An example of a cladogram



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(b) **Phenetic classification:** This is based upon evidence presented by numerical taxonomists who considered masses of data relating to present day morphological, cytological and biochemical similarities between organisms. A family tree based upon phenetic classifications is called a **dendrogram**.

3.8. Specimen identification and keys

Organisms are identified using keys. The key includes observable morphological characteristics of the organisms, which can be marched with those features which are diagnostic to a particular taxon. Such features include shape, color, number of appendages, segments, etc.

Identification essentially relies on appearance (phenotype) of the organism and it is therefore **artificial** and **phenetic**. There are various types of diagnostic keys the simplest is called a **dichotomous key**. The key is made up of pairs of statements called **leads**, numbered 1, 2, 3 and so on. Each lead deals with a particular observable characteristic. The paired statements of each lead should be contrasting on mutually exclusive. You follow the lead until the unknown organism is identified at the lowest taxonomic group possible. After each statement there is a number referring to the next lead to be considered. If the statement described in the lead matches the appearance of the specimen, the number following the statement indicates the next lead to consider.

Example: Key to microscopic organisms (from APHA, 1994).

- | | | |
|---|---|----------------------|
| 1a) Specimen of mass of filaments, requiring microscopic examination to determine details of structure | 2 | |
| 1b) Specimen a well organized nit or colony | 3 | |
| | | Algae |
| 2a) Cells contain one or more pigments | | Cyanobacteria |
| 2b) Cells colorless, lacking chlorophyll a | | |
| 3a) Organisms plantlike. Flowerlike structures, if present, do not respond when touched, generally are colored some shade of green, brown or red... | 4 | |
| 3b) Organisms animal-like, usually responds rapidly | | when touched, |



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whether attached or free living.....	5
4a) Plant structure relatively simple. Attachment structures may be present, but no true roots or fibrous tissue	Larger algae
4b) Plant structure usually includes true roots, stems and leaves	Higher plants
5a) Internal backbone present (vertebrates)	8
5b) No internal backbone present.....	Microinvertebrates
6a) Side appendages, if present, are fat lines.....	Fishes
6b) Side appendages, if present, are footlike, with separate digits.....	Amphibians

5.8 Summary

1. Micro-evolution is gradual evolutionary changes in populations that are built upon many small genetic variations that arise and are passed on from one generation to the other .
2. The shifting of gene frequencies in populations can be described using the Hardy_ Weinberg law.
3. Evolutionary changes occur when the Hardy-Weinberg equilibrium is upset either because mutation occurs, reproduction is nonrandom or the population is small.
4. Macroevolution refers to the evolutionary development of taxonomic units at the species level and above
5. Plants and animals are classified according to the binomial system of nomenclature in which the scientific name consists of only two words; the genus and the species name.

Questions

1. Differentiate between microevolution and macroevolution.
2. State the condition under which the Hardy-Weinberg equilibrium is upset.
3. Using your own example, illustrate the systematic placing of organisms onto the various taxa
4. Explain the following terms briefly:



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- i. Taxonomy
- ii. Binomial nomenclature
- iii. Cladistic

taxonomy LESSON 6

BIOLOGICAL TECHNIQUES

6.1 Introduction

Man, by his very nature is an investigative animal, which continually looks for information from nature. From the time we were toddlers to the time we became college students, we have been involved with varying levels of experimentation. We began by holding things with our feeble hands until we owned them as ours. In college, we have started searching answers for more intricate questions, requiring a clear scientific method. In this unit, we will learn about how to proceed with a scientific method and one of the common biological instruments_ the microscope.

6.2 The scientific method.

Science starts with formulation of a problem. This requires a highly creative imagination to come up with the right question. Answers must then be sought. The facts arising from the scientific investigation must be testable and repeatable by other scientists. It is essential that all scientific investigations be described fully and clearly. If investigations yield identical results under identical conditions, then the results may be accepted as valid.

_Some common terms used in scientific method:

a) Scientific observations

Observations may be of two types;

- I) Qualitative_ describing such characteristics as color, shape, taste, presence, absence, etc.
- II) Quantitative_ This is a more precise form of observation and involves the measurement of an amount or quantity, which may have been demonstrated qualitatively.

b) Hypothesis

A hypothesis is an assumption or question based on the observations that may provide a valid explanation of the observation.



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Albert Einstein stated that a hypothesis has two functions:

- i) It should account for all the observed facts relevant to that problem.
- ii) It should lead to the prediction of new information. New observations

(fact data) which support the hypothesis will strengthen it. New observations, which contradict the hypothesis, must result in it being modified or rejected.

c) Variables

In order to assess the validity of a hypothesis, it is necessary to design a series of experiments aimed at producing new observations, which will support or contradict the hypothesis. In most hypotheses there are a number of factors, which may influence the observation: these are called variables. The experiments vary one variable and observe the results of the variations.

- d) **Control experiment:** a control is defined as an experimental unit that has been given no treatment. Thus a control is usually the baseline against which the other treatments are to be compared. If an experiment has no control, it is impossible to conclude anything definite about the experiment.
- e) **Theory** A theory emanates from a hypothesis that has withstood attempts of falsification and continues to be successful in predicting previously unexplained facts and relationships. When a theory has proved invariable under all circumstances, or such variations as occur are systematic and predictable, then it may be accepted as a **law**.

6.3 Drawing of Biological Specimens

It is very important to record biological specimens through drawing. Drawing has the following purposes:

- I) To provide a record of work for future reference.
- II) To supplement observations and to enable you to see more fully and accurately the studies that you are investigating.
- III) To aid memory of what you see by actively recording.

In making your drawings, the following apparatus are required;



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- i) A notebook of suitable thickness and quality. ii)
- A sharp HB quality pencil. No colored pencils should be used.

The drawing should have the following qualities;

- I) They should be; large to include all parts are visible.
- II) Simple- to include the outline of the structure and other useful landmarks to show topography and relationships of parts.
- III) Should be accurate.
- IV) Lines should be sharp and clear_ each line should be considered and then drawn without removing the pencil from the paper. Shading and color should be avoided.
- V) Should be well labeled. There should be as complete as possible with label lines that do not cross. The labels should be spaced around the figure.

In making the drawings, the following rules should be followed;

- I) Two drawings should be drawn if necessary to include (a) a simple drawing of he main features and (b) another including details of small parts only.
- II) Draw what you see and not what you think you see, and certainly not a textbook copy.
- III) Every drawing should have a title, magnification, viewpoint of the specimen (such as TS) and explanatory notes.
- IV) Drawings of apparatus should be as vertical section showing clearly where there can be a flow of gases from vessels, through tubes and valves.

6.4 Reporting an experiment

All experiments should include the following in he report.

1. Title: This should be a clear statement outlining the problem to be investigated.e.g Experiment to investigate he digestive enzymes of a rat. It should be broad made specific by the hypothesis. short and precise.
2. Hypothesis or aim: This is a statement of he problem or the posing of a question. It may include an indication of the variables under examination and the possible outcome of the investigation.

Example:



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To investigate the effect of solutions of pH 2_10 on the rate of digestion of the protein albumin by the enzyme pepsin and to determine the optimum pH of he reaction.

3. Methods and procedure.

This is an account of the activities carried out during the performance of the experiment. It should be concise, precise and presented logically in he order in which the apparatus was set up and the activities performed during the experiment. It should be written in he past tense and not in the first person using information given, another scientist should be able to repeat the experiment.

4. Results and observations

This may be qualitative or quantitative and should be presented as clearly as possible in some appropriate form or forms, such as verbal descriptions tables of data, graphs, histograms, barcharts, kite diagrams, pie graphs etc. Where more than one measurement of he same variable is taken, the mean (x) should be calculated and presented.

5. Discussion

This should be brief and take the form of the answers to possible questions posed by the hypothesis or confirmations of the aim .The discussion should not be a verbal repetition of he results, but an attempt to relate theoretical knowledge of the experimental variables to the result obtained.

6. Conclusion:

A conclusion is included if there is a clear-cut verification of aim e.g.

There is a relationship between pH and enzyme activity and for this Reaction the optimum pH is x.

6. Reference:

This should include a full list of all cited literature material e.g. scientific publications, textbooks etc.

6.5 The Microscope

The microscope is a very common instrument in a biology laboratory. Its value lies in the following;

- a) It gives a magnified image of the object that can be identified.



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- b) It reveals the fine internal and external details of a structure or of small organisms.
- c) It takes measurements of the size and shape of the magnified object.

d) It takes various physical measurements in the analysis.

Microscopically techniques enable us to magnify objects up to 1000 times (also written as x1000) and thereby observe organisms as small as 0.5 micrometers.

6.5.1 Handling the microscope

The microscope is very expensive and care must be taken when handling it.

1. Keep the instrument in a box (or under a cover) when not in use in order to maintain it in a dust free state.
2. Remove it from the box when using and place it on the benches gently to avoid unnecessary jamming.
3. The lenses must be kept clean by wiping with a lens tissue.
4. The microscope must always be focussed upwards, never downwards. It is very easy to pass through the plane of focus when looking through the microscope and focussing downwards, and as a result damage the slide.
5. Keep both eyes open and use each eye in turn.

Things to do practically

1. How to focus under low power.
2. How to focus under high power.

6.5.2 Magnification:

The magnification of the object is the multiple of eyepiece and objective lens magnification.

Magnification of the microscope

Objective lens

Eye piece lens

Magnif. of the object.



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X10 X6 X60 X40 X6 X240 X10 X10 X100 x40 x10
x400

6.5.3 Oil immersion

For higher magnification than normal high power work (x400) an oil immersion lens can be used. The light gathering properties of the lens is greatly enhanced by placing a lid between the objective lens and the cover slip. The fluid must have the same refractive index as the lens itself so that the fluid used is generally cedar wood oil.

1. Place the slide on the stage and focus as normal for high power work.. Replace the objective lens with an oil immersion lens.
2. Place a drop of cedarwood oil on top of the glass cover_slip over the top of the object to be examined.
3. Focus the object again under low power, then swing the oil immersion objective lens so that the tip is in contact with the oil.
4. Look down the microscope and very carefully adjust the lens using the fine adjustment. Remember that at the plane of focus the lenses are only 1mm from the coverslip of the slide.
5. After use, clean the oil from the lens with soft tissue.

6.5.4 Types of microscopes

(a) The compound microscope

It has basically two parts; the stand and the optical system ranges from the traditional type with a curved “limb” to the more modern types with a curved or angled limb. In the traditional type, focussing is achieved by raising or lowering the body in relation to the stage.

In the modern type, focusing is usually achieved by moving the stage relative to the limb. This has the advantage that the controls are low down and are more convenient to handle than those on traditional stand, where they are usually mounted high on the limb.

The more traditional type of instrument is usually easier to maintain and to service.

(b) Inverted microscope.



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This is a variant of the above types in which the light sources and the specimen is mounted above the objective.

© Phase contrast microscope.

This is a modification of the light microscope that enables one to observe living or unstained preparations. This is made possible by two special devices that are fixed on the microscope: an annular disc which is placed beneath the condenser and an objective. It is made up of a glass plate, which is blacked out except for an annulus, which is left clean and through which the light rays can pass.

The annular disc and the objective lens fitted with a phase plate convert minute phase changes in transmitted radiations (due to differences in refractive indices of cell organelles) into perceptible shades of living cells become visible with fair contrast. It is very useful for studying living material and is only effective with objects which have little or no color of their own. Stained preparations don't really benefit from this technique.

Questions.

1. What is the key components of a phase contrast microscope?.
2. What is the principle reason for employing the phase contrast system?.

.....

e) The Electron microscope (also called Transmission Electron Microscope)

It is a large instrument consisting of a desk sized console and a column, which is the microscope proper that may be from about five to eight feet tall. Has ability to magnify 500 times better than a light microscope. Essentially the working principle of the two is the same. Both rely on a source of energy which is focused on and through a thin prepared specimen, and upon the formation of a magnified image by an objective system which is further magnified by a viewing arrangement.

There are however significant differences:

1. The electron microscope can give greater magnification because it uses a beam of high-energy electrons instead of a beam of light. The average wavelength of visible light is about



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550nm. The maximum possible resolution obtained with a high microscope is equal to half the wavelength of the light used approximately 200nm.

Objects smaller than this cannot be resolved. Many cells structures e.g. mitochondria, Golgi bodies etc are actually smaller than this.

2. The wavelength of electrons depends upon the voltage used to generate the electron beam, but in practice a resolution of about 0.5nm can be obtained, about 500 times better than the light microscope. The electron beam is produced by a device known as an electron gun apparatus, which is, housed at the top of the column of the electron microscope.

A high voltage (such as 50000 v) is passed through tungsten filament at the top of the column, and the filament emits a stream of electrons.

In order to focus the electrons (bend their path) electromagnets must be used rather than glass lenses.

The inside of the column is under a high vacuum to minimize both electrons scattering through collision with air particles and the subsequent heating that would occur.

In the transmission electron microscope electrons are focussed through the specimen. Only very thin sections or particles can be observed because electrons are easily scattered or absorbed by the object. Portions of the specimen with high relative molecular mass cause most scattering, so heavy metals such as lead and uranium are used in the staining of specimen to increase contrast.

The specimen is usually supported on small copper grid (about 2mm in diameter) which may be covered with a thin plastic film for extra support. After passing through the specimen, electrons are collected and focused by further electromagnetic lenses.

Electrons cannot be observed with the human eye and so they are either focussed onto a fluorescence screen, which produces a visible image. Or allowed to pass directly onto a photographic film or a permanent photograph called the **electron micrograph**.

Revision questions

- 1) What is meant by the term resolution?
- 2) What is the magnification of a microscope fitted with a 5 objective and a 6-eye piece?
- 3) Compare and contrast the operating principle of the compound microscope with that of the transmission electron microscope?



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4) What are the major structural differences between a light microscope and an electron microscope.

Answers

1. The resolution of an optical system is expressed as the distance between two points which are just far enough apart to be seen as two points. If they were any closer together they would be seen as one point only.
2. 30.
3. In answering the first part of the question you will need to dig out the salient facts concerning the operating principles of the two basic systems and write about what they have in common and where they differ markedly. Do not attempt on advantages or disadvantages table. The instruments are used for purposes too diverse for this.
4. Refer to the figure of the electron microscope and text.



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LESSON 7

KINGDOMS OF LIFE

Living things can broadly be divided into five categories or kingdoms:

- 1) Monera.
- 2) Protista.
- 3) Plantae.
- 4) Fungi.
- 5) Animalia.

7.1 Kingdom Monera

Organisms belonging to the kingdom monera were probably the first to appear on earth. This kingdom includes organisms that have prokaryotic cells, i.e. lacking a nuclear membrane separating DNA from cytoplasm, and without other membrane bound organelles. All the kingdoms are made up of organisms with eukaryotic cells i.e. have a nuclear membrane surrounding DNA (or genetic material), and with various membrane-bound organelles.

Organisms in the kingdom monera include bacteria and the blue-green algae (cyanophyta). Members of the kingdom monera are very small in size, and their internal structures have been studied with the advent of the electron microscope.

Among the prokaryotes, bacteria are the oldest and most abundant group in the world. Some bacterial fossils date from 3.5 billions years ago, long before the evolution of cells recognizable as eucaryotes of the modern type.

Most bacteria are heterotrophs ie they obtain heir energy from the tissues or body fluids of other living organisms. Some bacteria are harmful to man, causing many diseases such as cholera, venereal diseases, diphtheria, whooping cough, tetanus, etc. others however are beneficial, such as those which cause decomposition, and in the process recycling of nutrients in the environment, some are useful in the industrial and manufacturing process. The blue-green algae (cyanophyta) are mostly unicellular; occurring either singly or in colonies



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embedded in a gelatinous matrix. Some, however, are filamentous, they are called blue-green because they contain no chloroplasts and their pigments are blue-green. They include such as *Spirulina*, *Oscillatoria sp.*, *Anabaena sp.*, *Lyngbya sp.*, *Microcystis sp.* etc.

7.2. Kingdom Protista

Members of the kingdom Protista have unicellular (one-celled) eukaryotic cells. They are believed to have evolved from the monera group by gradually evolving various specialized organelles, such as chromosomes composed of nucleic acids conjugated to proteins and organized into microscopically clearly visible filaments, nucleoli and a nuclear membrane bounding off a definite nucleus.

The step from the prokaryotic (monera) to the first eucaryotes (Protista) was one of the big evolutionary transitions, second only to the origin of life. The question of how it came about is a matter of debate. The hypothesis suggests that larger, and more complex cells evolved when certain prokaryotes took up residence inside other prokaryotic cells. The hypothesis postulates that at about 2.5 billion years ago, oxygen began slowly to accumulate in the atmosphere as a result of the photosynthetic activity of blue-green algae. Those bacteria that were able to use oxygen in ATP production gained a strong advantage, and so such forms began to prosper and increase. Some of these evolved into modern forms of bacteria. Others according to the hypothesis moved into larger cells and evolved into mitochondria.

Eucaryotes differ from prokaryotes in that eucaryotes can undergo mitosis and meiosis. Mitosis allows the nucleus to duplicate its chromosomes precisely and to distribute them equally to two daughter nuclei. Meiosis is necessary to sexual reproduction, which also involves fertilization. The ability to reproduce sexually was an extremely important development because it allowed for genetic variation by the reshuffling of existing genes, in addition to the genetic variation reproduced by mutation to form new genes. Genetic variability is the raw material for natural selection, and so for evolution. The fossil record indicates that since their origin, eucaryotes evolved much faster than prokaryotes, giving rise to many different organisms with differing sizes, shapes and life spans.



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Included in the kingdom are the phyla Rhizopoda, Foraminifera, Zoomastigina, Apicomplexa, Paramyxea, Ciliophora, Euglenida, Ascomycotina, Acrasiomycota and Myxomycota.

7.2.1 Protozoa: These are the single-celled, mobile, heterotrophic protists that lack cell walls and have ingestive metabolism, e.g. *Paramecium* and *Amoeba*. They may be free living predators or scavengers, ingesting other organisms or bits of organic matter or parasites or multicellular symbionts. Parasitic protozoa infect more than ¼ of the world's human population. They also cause diseases of other animals and plants.

Characteristics of protozoa

1. They exchange gases with the environment by diffusion through the plasma membrane. Many live anaerobically when there is no oxygen.
2. They have contractile vacuoles which collect excess water and then contract, discharging the water.
3. Under adverse conditions, they form cysts which are resistant to desiccation and temperature changes. Many protozoa are dispersed as cysts.
4. Many protozoa are motile, moving by cilia, flagella, pseudopodia or by flexing the cell body.
5. Protozoa are either heterotrophic, ingesting food by endocytosis or absorb organic molecules from their environment.
6. Most protozoa can sense stimuli, e.g. light, temperature, touch, gravity and chemicals.
7. Both asexual and sexual reproduction occur in protozoa. Asexual reproduction occurs by mitotic division. The asexual reproductive mechanisms include binary fission, multiple fission and budding. Sexual reproduction occurs by syngamy or conjugation.

Several phyla fall under protozoa:

- (a) **Phylum Zoomastigina (Zooflagellata):** They have flagella, either one or many. They may be free living, either in marine or fresh water while others are symbionts or parasites, e.g. *Trypanosoma gambiense* which lives in the bloodstream of vertebrates, causing African trypanosomiasis. Parasitic forms have life cycles involving two hosts.



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Choanoflagellates are free living, either solitary or colonially and are sessile. They feed by filter feeding.

(b) **Phylum Sarcodina (Rhizopoda):** They have pseudopodia for feeding and movement e.g. *Amoebae proteus*. They are mostly free living but some are parasitic. The naked amoebas are asymmetrical and possess relatively few organelles. Their cytoplasm is divided into an outer ectoplasm and an inner endoplasm. They lack flagella but when present, are restricted to developmental stages. Reproduction occurs asexually by fission and sexually.

(c) **Phylum Ciliophora:** They are also called ciliates. They have cilia all over the body. An example is *Paramecium*. Ciliates may have trichocysts that are used for anchorage, defense and prey capture. Sometimes they have an external cup or **lorica**, which is protective. A unique feature of ciliates is that they have two different types of nuclei, a micronucleus which is involved in sexual reproduction, and a macronucleus which controls the cell's growth and metabolism. The contractile vacuoles, which are typically two, discharge excess water at the cell surface. Most ciliates prey on bacteria, small animals or fellow protists. Some eat organic particles from the surrounding water. Some are symbionts while a few are parasitic. Reproduction occurs by asexual and sexual reproduction. Sexual reproduction occurs by conjugation.

(d) **Phylum Apicomplexa**
This phylum arose by splitting the Sporozoa into Microspora and Apicomplexa. They are all parasites and have life cycles with two different host species. One stage has an apical complex, a distinctive arrangement of organelles and cytoskeletal fibres at one end of the infective, motile stage. These are used to penetrate the host cells. They lack cilia and flagella, except in microgametes of some species. Sexual reproduction occurs by syngamy while asexual division occurs by schizogony. Examples include the malaria parasite, *Plasmodium*.



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(e) **Phylum Microspora** e.g. *Nosema*

They are all parasites and have unicellular spores. They are obligatory intracellular parasites with plastids (particles in the cytoplasm containing pigments, starch, proteins and oil).

Algae

Some algae are grouped within the kingdom Protista. These include;

- Euglena (Euglenophyta)
- Diatoms and golden algae (Chrysophyta or Bacillariophyta)
- Dinoflagellates and related algae (Pryophyta or Dinoflagellata).

Most of these algae are photosynthetic and belong to a group of algae called phytoplankton. Phyto means plant; plankton means wanderer-- that is, they float near the surfaces of oceans, lakes and ponds. They are ecologically important as the ultimate source of food for aquatic animals. They also produce an estimated 30-50% of the oxygen in the atmosphere.

7.3 Kingdom Fungi

Fungi are like plants in that they have cell wall, but they lack chlorophyll. The cellwall of higher plants is made up of cellulose that of fungi is made up of chitin. Fungi are heterotrophic, that is to say, they absorb their food particles from their surroundings. The storage product of fungi consists of oil and glycogen, which is similar to the storage products of animals. For this reason, fungi have been placed into their own kingdom. Examples include Zygomycotina e.g. *Rhizopus* (black bread mold)

7.4 Kingdom Plantae

All plants have rigid cellwalls and subcellular membrane-bound structures containing chlorophyll. They synthesize their own food, through photosynthesis, using inorganic substances obtained from the soil. Unlike prokaryotes, plants are eucaryotic and multicellular; ie they consist of more than one cell, with some cells being specialized to some degree to perform different tasks.

7.5 Kingdom Animalia.



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The animal kingdom belongs to the group commonly referred to as metazoa which by definition have the following characteristics.

- 1) Animals are multicellular and eucaryotic. In being multicellular, animals have cells differentiated into tissues, and most have tissues aggregated into organs.
- 2) Animal are heterotrophs except for few that absorb small organic molecules directly from host tissues e.g. tapeworms, animals ingest particles or chunks of food which are then broken down (digested) and absorbed.
- 3) Animals reproduce sexually, with large, non-motile eggs and small flagellated sperms. There are however many exceptions to its rule, for some animals reproduce asexually through parthenogenesis.
- 4) Animals go through progressive stages of embryonic development.
- 5) Animals are motile during at least part of the lifecycle.
- 6) They mostly require oxygen for respiration.

Animals are believed to have arisen evolutionarily from ancestral protists through coloniality, a process of division of labour among many cells.

Animals are divided into several phyla, which include:

- (a) Phylum Porifera (sponges)
- (b) Cnidaria (coelenterata)
- (c) Platyhelminthes (flatworms)
- (d) Nematoda (roundworms)
- (e) Mollusca (shell animals)
- (f) Annelida (segmented worms)
- (g) Arthropoda (arthropods)
- (h) Echinodermata (spiny-skinned animals)
- (i) Chordata (animals with backbone)

LESSON 8

LOWER INVERTEBRATES



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8.1. Phylum Porifera (the sponges)

Most sponges live in shallow ocean waters, although a few are freshwater.

Adult sponges are sessile and live attached to the bottom of clear water. Their bodies are simple, vase-shaped with a central cavity opening (osculum) at the top (Fig. 9.1). Sponges have no organs. Not even a digestive system. They have no coelom. They have cells aggregated into water canal systems. Water moves through the tube-like cells into the spongocoel, then out through the osculum. Movement of flagellum possessed by the choanocytes facilitates water movement. Oxygen and nutrients in water is absorbed by the collar cell microvilli. The sponges feed by filter feeding, facilitated by the choanocytes. Freshwater forms have contractile vacuoles to get rid of excess water that enters their bodies.

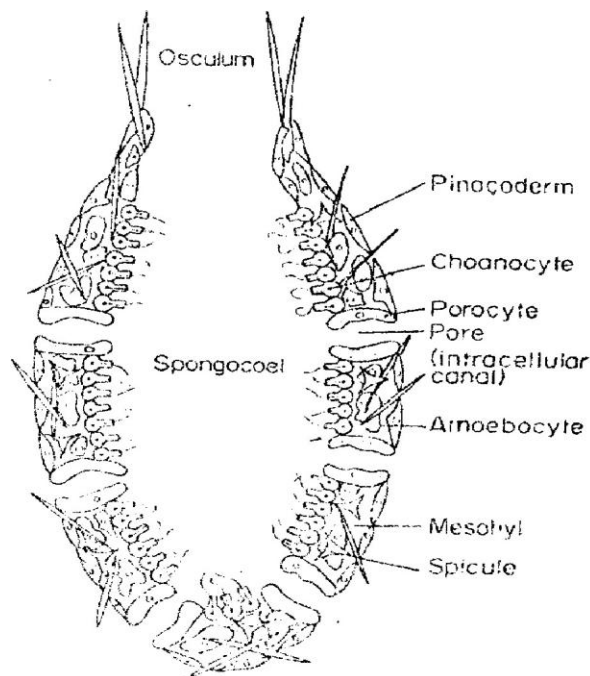


Fig. 8.1: An asconoid sponge body plan

Sponges are usually hermaphroditic, with each individual producing both sperm and eggs. Sperms are released into water and travel to the eggs. The fertilized eggs are either released and develop on their own, or they develop to the larval stage inside the parent.



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Sponges also reproduce asexually by budding, formation of reduction bodies and gemmule formation. A gemmule is an aggregate of food-filled amoebocytes and other cells, often enclosed in a hard covering. Other asexual reproductive mechanisms are regeneration and fragmentation.

The phylum has 4 classes:

Class Calcarea (e.g. *Sycon*)

Class Demospongiae (e.g. *Spongilla*, *Spongia*)

Class Sclerospongiae

Class Hexactinellida (e.g. *Hyalonema*, *Euplectella*)

8.2 Phylum Cnidaria

The cnidarians include hydra, jellyfish, corals and sea anemones. Most have tentacles around the oral end armed with stinging cells (nematocysts). The nematocysts are used for offense and defense and prey capture. All have radial symmetry. The body wall is three layered, with an outer epidermis, an inner gastrodermis and middle mesoglea (Fig. 9.2). There is a single cavity, the gastrovascular cavity, which serves both to digest food and to distribute food around the body. Cnidarians are carnivorous and either capture prey using their tentacles or are suspension feeders. Cnidarians have two body forms, a polyp and a medusa. The polyp is sessile while the most medusas are free swimming and is the sexually reproducing form. In sexual reproduction, gametes form from the interstitial cells, epidermis, or gastrodermis. The gametes are freed into the gastrodermis and emerge through the mouth. Cnidarians may be either monoecious or dioecious and fertilization may be external or internal.

Asexual reproduction occurs by budding, fission and fragmentation.

The phylum Cnidaria has three classes:

- I. Class Hydrozoa (hydra) which are fresh water and obelia which is marine.
- II. Class Scyphozoa (jellyfish, *Aurelia*)
- III. Class Anthozoa (stony corals and sea anemones)



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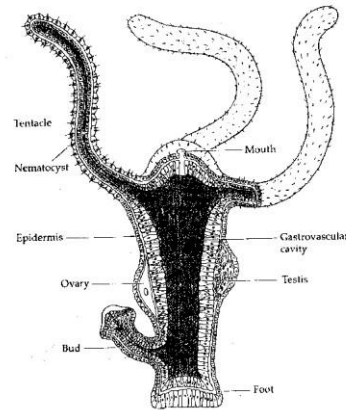


Fig. 8.2: Body form of a hydroid polyp

8.3 Phylum Platyhelminthes (flat worms)

Platyhelminthes includes free-living tubelarians, parasitic flukes and parasitic tapeworms. All are bilaterally symmetrical i.e. have a definite anterior and posterior ends orientating to their environment. Flatworms however, lack special respiratory and circulatory structures. They depend upon diffusion for their oxygen needs and their flattened structure is one way to keep all internal cells close to the surface. The digestive system is also incomplete (pouch-like rather than tubular), with a highly branched cavity. They have no body cavity (are acoelomate) and are soft bodied, with protective mucus secretions while some have internal spicules. Flatworms feed almost exclusively from other animals and are generally hermaphrodite.

The phylum is divided into 3 classes:



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(a) **Class Turbellaria** (planarians): These are free living, ciliated planarians. They are hermaphroditic with well developed reproductive systems utilizing fertilization. They can also reproduce asexually by transverse fission. Large species move by muscular undulations and mucus covering their bodies lubricates and protects the body surface, forms trails and can be used to entangle prey. They have sense organs such as eyes and ciliated patches which may be important in food location.

(b) **Class Trematoda** (flukes)

This class includes the flukes. They are similar to planarians but are not ciliated. Flukes are common internal parasites, many of them causing diseases of man and domesticated animals. Their life cycle involves a primary host (a vertebrate) and between 1-3 secondary hosts, one of which is a snail. They have attachment organs consisting of an oral sucker and a mid-ventral sucker (acetabulum). The tegument plays an important role in protecting them against the host enzymes and antibodies and in absorption of food.

An example is the human fluke, *Schistoma mansoni*, whose infestation causes schistosomiasis. *Schistosoma mansoni* is common in irrigation canals and enters the body through the skin as people wade through the water.

© **Class Cestoda** (tapeworms)

These are the tapeworms, which are endoparasitic usually in the intestines of vertebrates. They lack digestive tracts, and absorb digested food from the intestine of their hosts through the tegument. The tegument also assists in digestion of food. The body of tapeworms consists of a scolex equipped with suckers and hooks, and a series of proglottids, which bud off the scolex. Cestodes have no sensory organs. They produce a large number of eggs and have rapid growth rates. They have anaerobic metabolism and so can adapt to the low oxygen content in the intestines. Cestodes are generally hermaphrodite, each proglottid containing both male and female reproductive organs.



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8.4. Phylum Nematoda (roundworms)

Nematodes have perfectly cylindrical shapes, and are usually tapered at both ends. They are bilaterally symmetrical, although from outside appearance, they lack a distinctive head. A resistant but elastic cuticle surrounds most.

Sexes are separate though hermaphroditic species also exist. Most are parasitic, but some thrive in nearly every conceivable moist or aquatic habitat, from the soil of flower gardens to the world's oceans. Nematodes feed with a muscular pharynx. They feed upon protozoans, rotifers small earthworms and also upon themselves. Some nematodes are parasitic on plants and animals. They have a pseudocoelom, which acts as hydrostatic skeleton. Hydrostatic skeletons consist of a incompressible liquid trapped in the compartments of a worm's body. It supports the animals against the pull of gravity and provides a firm body part against which muscles can pull. The skeleton provides for movement through the host tissues and fluids and interstitial spaces in sediments. Although sexual reproduction occurs, some species reproduce by parthenogenesis and gynogenesis.

Examples of parasitic nematodes include *Ascaris lumbricoides* and *Wuchereria bancrofti*, which causes elephantiasis (filariasis).

Nematodes produce a large number of eggs (sometimes close to 30 million) having resistant egg cases, which pass out with faeces. They have low oxygen requirements.



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LESSON 9

HIGHER INVERTEBRATES

9.1. PHYLUM ANNELIDA

These are segmented worms. This phylum is represented by the earthworms, *Lumbricus* sp. They are found in marine, freshwater worms and terrestrial habitats.

The Annelids are worms whose bodies are divided into similar rings or segments, arranged in linear series and externally marked by circular grooves called **annuli**. The body segmentation is commonly referred to as **metamerism**. Metamerism also occurs in the arthropods (insects, crustaceans and others), which are related to the annelids. They have a coelom, which provides a hydrostatic skeleton. The circulatory system is closed, blood flowing in closed vessels. Reproduction may involve copulation but asexual reproduction also occurs. Respiratory exchange occurs across the extensive body surface, whose surface area is increased by the thin, flattened parapods. Excretion of nitrogenous wastes occurs through nephridia.

Annelids are at times called bristle **worms** because with the exception of the leeches, most annelids bear **tiny chitinous bristles called setae** which help to anchor the somite during locomotion to prevent backward slipping. Aquatic forms have long setae, which help them in swimming.

Many Annelids live in burrows or secreted tubes. The stiff setae help them in preventing the worm from being pulled out or washed out from its home.

Annelids are divided into three classes:

- i) Polychaeta
- ii) Oligochaeta
- iii) Hirudinea



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9.1.1 Class Polychaeta (Bristle worms)

This class includes the ringworm *Nereis*, Lugworm, *Arenicola* and the palolo worm *Eunice*.

The bristle worms are so called because they have many bristles that protrude from their bodies. Most are marine, though some live in the brackish and freshwaters. A few species are terrestrial. Along the sea shore, polychaetes are the most abundant animals, but because they live under stones, in burrows, in mud or sand, they are only seen by intriguing biologists, or by people who dig them up to use as fishing bait.

Polychaetes may be herbivores, predatory carnivores, scavengers, deposit feeders or filter feeders. They have strong jaws, used to seize food. They have separate sexes and reproduction occurs by sexual and asexual means. Asexually, they reproduce by budding. Epitoky is the formation of a modified sexual form called an epitoke that differs in many respects from the atoke or non-sexual form (Fig. 10.1).



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Fig. 10.1: A male polychaete budding off a chain of epitokes

Activity

How does an epitoke differ from an atoke?

9.1.2. Class Oligochaeta.(e.g *Lumbricus sp.*)

The earthworms are the most familiar oligochaetes. They have no parapodia and possess few, inconspicuous setae, hence the name of the class. Their head is reduced compared to the polychaete head and has no sensory tentacles. They burrow in moist, rich soil, emerging at night to explore their surroundings. In dump, rainy weather they stay near the surface, often with mouth and anus protruding from the burrow. In very dry weather, they may burrow several feet underground, coil up in a slime chamber, and become dormant.

Earthworms move by **peristaltic movement**. This is achieved by contraction of circular muscles in the anterior end, which then lengthen the body. This pushes the anterior end forward where it is anchored by setae. Contraction of longitudinal muscles shortens the body, pulling the posterior end forward. As these waves of contraction pass along the entire body, it is gradually moved forward.

Earthworms are monoecious (hermaphrodites) that is both male and female found in the same animal. They have a clitellum which is an area of epidermis swollen with unicellular glands. It produces mucus for copulation, for secreting the wall of the cocoon and for secreting albumin in which eggs are deposited within the cocoon. Hermaphrodites are thought to be an adaptation to any mode of life, which produces relatively few contacts between individuals. Immature sperms are released from the testes into the seminal vesicles, where they mature, and then pass on into sperm funnels and down sperm ducts to the male genital pores in somite 15, where they are expelled during copulation.



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9.1.3 Class Hirudinea (Leeches)

These are predominantly a freshwater group and are found in lakes, sluggish streams, ponds, marshes or open ditches. Leeches are very similar to oligochaetes, and are probably derived from them, except that they have had little success in meeting terrestrial stresses.

Most leeches are ectoparasites, sucking blood of human beings and other aquatic vertebrates. Some however are scavengers or predators on small invertebrates such as earthworms, insects, slugs and snails?

They have no larval stage, no distinct head, are hermaphroditic with cross-fertilization and eggs are laid in cocoons. They also possess a clitellum and have anterior and posterior suckers. They differ from oligochaetes in that they lack setae, the coelom is reduced and filled with connective tissue and has no septa.

9.2 Phylum Mollusca (shell animals)

9.2.1. Introduction

The Mollusca is one of the largest animal phyla after the Arthropod with nearly 50000 living species. The name mollusca indicate one of their distinctive characteristics, **soft bodies**. The molluscs range in size from minute to over 18mm long. Among their common characteristics include:

1. A mantle: The mantle is a fleshy covering that produces external shells through secretions.
2. Foot: All have a muscular foot that contains sensory and motor systems and may be used for creeping, digging, holding on, and capturing prey.
3. Gills: Molluscs have feathery respiratory gills.
4. Shell: All have external shells secreted by the mantle and lined by it.
5. Most molluscs are dioecious, although some are hermaphroditic.

The life histories of molluscs suggest a relationship of Mollusca to Annelids.

1. Firstly, many primitive molluscs, like primitive annelids, have an egg that hatch into free-swimming larvae called trochophore. The trochophore then metamorphoses directly into a small juvenile.
2. Some fossil molluscs were segmented and apparently had a well developed coelom,



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similar to that of animals.

The phylum Mollusca is divided into several classes:

9.2.2 Class Amphineura (Polyplacophora) (the chitons)

Chitons are among the least specified molluscs. They have a flattened body covering a long foot. The body consists of a chain of 8 dorsal shell plates for flexibility, so that they can roll into a ball. They are most common along the coast. They prefer rocky surfaces, where they creep out only very short distances for feeding. They feed by scraping algae from the rocks, using their large radula. When disturbed, their defense is to grip the rocks with powerful foot muscles and flatten so that it is very difficult to remove them.

Figure of a chiton.

9.2.3 Class Gastropoda (the snails)

Gastropods form the largest and most successful group among the molluscs. They contain about 35000 living and 15000 fossils species. The group includes:

Snails

Limpets

Slugs

Whelks etc.

Gastropods occupy a wide range of habitats, being common in the littoral zones of seas and at great depths. Some are even pelagic (occurring in the water column). Some are adapted to brackish water and others to fresh water. On land they are restricted by such factors as the mineral content of the soil and extremes of temperatures, dryness and acidity. Even so, they are widespread where they occupy all kinds of habitats including ponds, pastures, woodland, under rocks, on trees, underground and on bodies of other animals. They have successfully undertaken every mode of life except aerial locomotion.

Land snails

Land snails have evolved from marine molluscs, some directly, some through freshwater ancestors. They have two subclasses: Prosobranchs and Pulmonates



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- i) Prosobranchs: These lack lungs but have an operculum, which closes the shell aperture when the animal is retracted. This group also contains most of the marine snails and some of the freshwater and terrestrial gastropods.
- ii) Pulmonates: Snails in these order lack gills but possess lungs. They lack the calcareous operculum for closing the shell, and usually both sexes are present in each individual.

Snails undergo **torsion** in the course of their development. The shell and the visceral organs undergo a 180° counter clockwise rotation relative to the foot. This produces the typical gastropod asymmetry. Before torsion occurs, the snail embryo is bilaterally symmetrical with an anterior mouth and posterior anus and mantle cavity. Torsion is brought about by an even growth of the right and left muscle that attach the shell to the headfoot. After the organism undergoes torsion, the anus and mantle cavity opens anteriorly above the mouth and head. Torsion also shifts the gill, kidney and other organs, originally on the left side to the right and the organs originally on the right side to the left side. It also bends the digestive tract and twists the nerve cord onto a figure eight. the anus comes above the head where fecal excretion occurs. The adaptive importance of torsion is that it creates space into which the head can be accommodated when it is retracted. The chemosensory sense organs (osphradia) can sense water ahead of the animal. However, there is fouling of the cephalic sense organs and it leaves less room for organs on the right side of the body of the mantle cavity (Figure 10.3).

9.2.4. Class Bivalvia (or Pelecypoda)

These include the mussels, clams scallops and shipworms. Most bivalves are sedentary filter feeders that depend on ciliary currents produced by the gills to bring in food materials. Unlike the gastropods, they have no head and very little cephalization. Most are marine, but may live in brackish water and in streams, ponds and lakes. Bivalves are laterally compressed, and their two shells (valves) are held together dorsally by a hinge ligament that causes the valves to gape ventrally. The valves function largely for protection. The foot is also laterally compressed and is used for burrowing.

9.2.5. Class Cephalopoda

This class includes the squids, octopuses, devilfish and cuttle fish. They are all marine and active predators. Cephalopods are the “head-footed” molluscs in which the modified foot is



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concentrated in the head region. The edges of the foot are drawn out into arms and tentacles that bear sucking discs for seizing prey. Part of the foot is also modified to form a funnel for expelling water from the mantle cavity. A squid relies on speed for protection. The mantle is thick and muscular is protective besides being the chief swimming organ. Generally, they have an ink sac from which they eject a cloudy substance into water as they move away from danger. This protects them from predators.

The circulatory system is closed, providing for a rapid distribution of oxygen through the tissues. Respiratory exchange is speeded up by the extensive gills and by blood pigments. Sense organs are numerous and large. At the rear end the upper surface is extended into a pair of triangular fins that stabilize the animal and can be adjusted to change direction of movement. Sexes are separate and one of the male arms is modified into a hectocotylus, used for transferring sperms to the female.

9.3. Phylum Arthropoda

The phylum Arthropoda contains three times as many species as all other phyla combined. The term arthropod refers to “jointed foot” which is an important characteristic of all animals within the phyla. They are bilaterally symmetrical, segmented animals with jointed appendages. The body surface is covered by a chitinous exoskeleton, which enables arthropods to live on dry land. The cuticle is tough and flexible. It is important for protection and support but also allows movement at the joints of the appendages. Organ systems are well developed but the coelom is generally reduced and forms the haemocoel. Sexes are generally separate.

The phylum has four subphyla which include: trilobitomorpha, chelicerata, uniramia and crustacean.

9.3.1 Subphylum Trilobitomorpha

These are the most primitive of all arthropodan groups and are extinct. They were abundant in the Cambrian and Ordovician eras and disappeared at the end of the Palaeozoic. The body was divided into three regions, with jointed biramous appendages on the ventral surface. They possessed a pair of compound eyes and a pair of antennae on the cephalon.



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9.3.2 Subphylum Chelicerata

Chelicerates live in shallow water along sandy or muddy sea shores. Their body consists of a cephalothorax and an abdomen. They characteristically have feeding appendages called cheliceras and limbs are biramous.

Class Arachnida: They are mostly terrestrial and include spiders, scorpions, ticks, mites harvestman, etc. The cephalothorax bears 6 pairs of appendages of which the first pair in front of the mouth are the cheliceras. These end in pincers or fangs. The next pair are the pedipalps, which are sensory (in spiders) or are used for seizing prey (scorpions). The next 4 pairs are walking legs. The abdomen does not bear appendages. They have no compound eyes, antennae or mandibles. Arachnids respire by means of booklungs and have open circulatory systems. Excretion occurs by means of Malpighian tubules.

Class Merostomata

These are the horse shoe crabs. They are aquatic and have gills on the abdominal appendages, for respiration. Their cheliceras bear pincers, which are used to grasp food. The walking legs are used to manipulate food. The flattened abdominal appendages are used for swimming and fast scuttling along the bottom.

9.3.3 Subphylum Uniramia (Mandibulata)

Members of this subphylum are primarily terrestrial though there are freshwater forms. They have uniramous appendages. The body is divided into head, thorax and abdomen. The head appendages include a pair of antennae, mandibles and maxillae. In contrast to the chelicerates, they lack chelicerae or chelate limbs. The exoskeleton may be calcareous or non-calcareous.

They have separate sexes and fertilization is internal.

9.3.3.1 Superclass Myriapoda: These arthropods have a hardened head capsule and a pair of antennae and mandibles.

Class Chilopoda (centipedes). They are predatory terrestrial arthropods having poison claws on the first trunk appendages. There are a single pair of appendages on the segments, except the last two.



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Class Diplopoda: (Millipedes). They are also terrestrial. They are herbivorous, feeding on decaying vegetation. Some members have calcareous exoskeleton. Some segments have 2 pairs of legs, heart ostia and

ganglia.

9.3.3.2 Superclass/Class Insecta (Hexapoda): These are mainly terrestrial, some being aquatic especially at larval stages. Insects differ from the myriapods in that the trunk is subdivided into a thorax having three segments and an abdomen. Insects have wings on 2nd and 3rd segments of the thorax, which also bears three pairs of appendages. They exchange gases through tracheae and have open circulatory systems. They have a variety of sense organs which include simple and compound eyes, chemoreceptors and sound receptors. Their mouth parts are adapted to a wide range of feeding habits, including chewing, piercing and sucking, sponging, etc. Examples are grasshoppers, crickets, termites, lice, bugs, water boatman, mayflies (aquatic larval stages), dragonflies, damselflies, butterflies, moths, beetles, caddis flies, mosquitoes, fleas, wasps, ants, bees, etc.

9.3.3.3 Superclass/Class Crustacea: The word crustacea refers to the flexible crust that covers the body of these arthropods. They are mostly aquatic with gills, 2 pairs of antennae, mandibles and 2 compound eyes, typically stalked. The body is divided into a cephalothorax and abdomen. A backward fold of the head, the carapace, is diagnostic of many orders. The appendages are biramous and are modified for various functions. The respiratory system is extensive and provided by gills, which are attached to the 2nd maxillae and 2nd to 7th thoracic appendages. The circulatory system is open and excretion is carried out by means of the green (antennary) gland. Sexes are separate. The class include shrimps, lobsters, crabs, barnacles, cladocerans, rotifers, copepods, ostracods.

9.4 Success of Arthropods

Why are arthropods so successful in terrestrial habitats?

The answer lies in the fact that arthropods are efficiently able to **waterproof** themselves by having a waxy chitinous jointed exoskeleton.

The skeleton also:

- i) Offers protection to the soft tissues within.
- ii) Offers strong, flexible framework upon which to operate the jointed appendages.



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Layers of the chitinous exoskeleton or cuticle are secreted successfully by the underlying epidermis. The cuticle is non-living but is hardened to offer protection and support.

In order to allow growth of the arthropod, the hard outer covering is periodically discarded and replaced with anew cuticle. Such a process is called **molting**. During muting, an aquatic arthropod takes large quantities of water in its digestive tract, blood and tissues. A terrestrial arthropod takes air into its digestive tract and respiratory system. The rapid distension of the animal by extra water or air ruptures the old cuticle, which is cast off, a process called **ecdysis**.

9.5 Success of Insects

Insects are the most successful among the arthropods. There are many factors that contribute to the success of insects, among which include:

1. Small size
2. Wings
3. Metamorphosis

Small size allows an insect to complete its whole life history by specializing in the use of a single limited resource. A single plant, for example, that would provide only a few bites for a large animal can support hundreds of small insects comprising many species, each of which takes advantage of some different plant part. Individuals of different species specialize in eating leaves, buds, flowers fruits bark or roots, others drink nectar or collect the pollen, and yet others suck the sap or feed on plant tissues in various stages of decay.

Wings permit insects to exploit specialized food sources that are rare or widely scattered, or to fly to different localities, as resources change with the seasons, and draw on several resources, any one of which by it, would be insufficient.

Metamorphosis, that's the transformation from egg -larvae -pupae-adult allow insects to have different eating habits that can profit from a combination of resources and habitats. For example, the larvae may feed on plant stamens, pistils and other flower parts, a food resource available for only a short time say 3 weeks, while adult may feed on leaves and buds.



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Revision questions.

1. Outline classifications of the following phyla

☐ Mollusca ☐ Annelida

☐ Arthropoda

2. Comment on the following unique features:

☐ Torsion in mollusca

☐ Metamerism in some invertebrates.

3. Write short notes on:

☐ chitons

☐ bivalves

☐ polychaeta

☐ hirudo

☐ crustaceans

☐ insecta

☐ diplopoda

☐ trilobitomorpha

LESSON 10 THE DEUTEROSTOMES

10.1. Phylum Echinodermata

These spiny-skinned animals have the following features:

- they have a skeleton of calcareous plates under the skin
- Water vascular system with tube feet.
- They have a penta radial symmetry with no head.
- They are all marine.
- Most of them have separate sexes.



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(a) **Class Crinoidea** (sea lilies and feather stars)

The class has
about 80 species

The characteristics include:

- They are either stalked or free
- They have branched arms with feeding grooves.
- Ciliated tube feet used for feeding and gas exchange.

(b) **Class Asteroidea** (sea stars)

- They are flattened with a central disc.
- Their mouth lies on the underside (oral surface) of the central disc.
- They move by their tube feet located on the underside of the body.
- Most are carnivorous and feed on snails, crustaceans, bivalves, and polychaetas fish, using their tube feet to gulp the prey. (“Crown of thorns” sea star feeds on cnidarian polyps from destroyed coral reefs).
- They can regenerate a whole body from a broken arm still attached is part of the central disc. This is why cutting them to pieces by fishermen does not help.

The water-vascular system

It consists of a system of interconnecting fluid-filled tubes or canals. There is a ring canal encircling the mouth and 5 radial canals running along the underside of each arm. From the radial canals extend tube feet or podia. Each tube foot is connected to a rounded muscular sac called the ampulla. When muscles in ampulla contract, water is forced into the foot and it extends. When muscles in footwall contract fluid is forced back into ampulla and radial canal. This lowers pressure in foot and it shortens.

(c) **Class Ophiuroidea** (Brittle stars and serpent stars)

- They resemble asteroids except their arms are sharply marked off from the central disc.
- Second difference with asteroids is that they move using the wriggling of the jointed arms rather than the tube feet. Some even use their arms to swim.
- The tube feet are used as sense organs and for delivering food to the mouth.
- Their arms are feathery due to the spines that cover them.
- They eat organic detritus from the sea floor.



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- They have a very simple gut with sac like stomach and no intestine or anus.

(d) **Class Echinoidea** (Sea urchins and sand dollars)

- They are tubby (short and fat) little echinoderms.
- They live mouth down on bottom of sea.
- They have brittle calcareous spines for protection.
- They have calcareous plates, which are fused into a sphere (and for sand dollar into a flattened envelope), pierced by holes for anus, mouth and tube feet.
- The mouth contains a chewing apparatus with a ring of large triangular teeth, part of the feeding mechanism known as ARISTOTLES LANTERN.
- Echinoids can chew all organic matter encountered.
- They walk either on their spines or tube feet.
- The tube feet are also used for gas exchange.

(e) **Class Holothuroidea** (Sea cucumbers)

- They resemble large flabby sausages.
- They lie partly buried in the sea floor.
- Their calcareous skeleton is reduced to microscopic plates scattered through the skin.
- They lack arms
- Their mouth is on one end so they lie on their side.
- They can move slowly either by use of their tube feet or by wriggling.
- Their mouth is lined with tube feet known as tentacles. These tentacles secrete a net of mucous used to catch plankton (food); others are detritus feeders.

(f) **Class Concentricycloidea** (Sea daisies)

- Have two concentric rings of the water vascular system
- Outer ring, not radial canals, bears tube feet and ampullas
- There is no internal digestive system. A thin velum can be applied o the surface of decomposing food to digest and absorb nutrients



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10.2 Phylum Hemichordata

These are the acorn worms. The phylum is divided into two classes: Enteropneusta and Pterobranchia

- They were originally classified as a chordate sub phylum on the mistaken belief that the collars connective tissue was a notochord.
 - It is now known that this is not true and the only other chordate features are the gill slits leading from the pharynx to the exterior.
 - The epidermis is covered with cilia.
 - Their body is divided into three parts;
 1. A prosomal proboscis, used for burrowing and feeding.
 2. A mesosomal collar, used for anchoring
 3. A metasomal trunk, which is the main body
 - Hemichordates have features in common with echinoderms and chordates:
 - (i) transparent hemichordate larva quite resembles an asteroid echinoderm
 - (ii) Pterobranchs have a hydrostatic system in their tentacles reminiscent of the water vascular system of echinoderms or of lophophorates system of tentacles.
 - Some enteropneusts have a dorsal tubular nerve cord.
 - The young of some species have a post anal tail, as in chordates
- The hemichordates thus appear to represent some evolutionary link between echinoderms and chordates; infact, some zoologists e.g. Libby Hyman suggested that hemichordates are a common ancestor of the two groups.



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LESSON 11

Phylum Chordata

11.1 Chordate Features

The five major features of chordates are:

- **Notochord:** This is a stiff and rod-like structure found in all chordates at some stage of their life. In the cephalochordates and urochordates, it serves as a skeleton and prevents the body from shortening when body muscles contract. In vertebrates it is replaced by column of vertebrae, the backbone, in fact the notochord exists only in the embryo.
- **Pharyngeal gill slits:** All chordates have gill slits at some time of their lives. Gill slits lead from the pharynx to the exterior it is the throat cavity behind the mouth.
- **Dorsal tubular nerve cord:** Chordates have this type of nerve cord, remember that invertebrates have nerve cords that are solid and ventral.
- **Segmented body:** Most chordates have segmented bodies, locomotion is by segmented blocks of muscle (myotomes) pulling on the internal notochord. It is this that allowed early chordates to swim quickly and efficiently by side to side wriggles of the body. As they swam they took water and food through the mouth and water then left through the gill slits having given up oxygen to the blood in the gill. Sense organs in the head detected where the animal was going to and the brain and nervous system became well developed.
- **Endostyle:** This secretes mucus, used to trap food.
- **Postanal tail:** At some stage of life a tail extends behind the anus at the posterior end of the body. In non-chordates the anus is usually at the end of the tail.

The phylum is divided into three subphyla: Urochordata, cephalochordata and vertebrata. The invertebrate chordates lack a vertebral column, live passive lives as filter feeders and are all marine. They constitute 3% of the phylum while the vertebrates constitute 97%.



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11.2 Subphylum Urochordata (tunicates)

- These are the sea squirts (ascidians) that are entirely marine. Adults resemble chordates only in possession of pharyngeal slits and in having an endostyle. They are classified as chordates because they have a tadpole-like larvae stage with a notochord and a hollow dorsal nerve cord. However the notochord disappears during metamorphosis to the adult form.
- Members of the class ascidiacea are sessile and secrete a test (protective outer covering of the polysaccharide cellulose). It is only in this group that one finds cellulose in the animal kingdom. Ascidians are mostly filter feeders. Many live in colonies and share a common mouth and common cellulose. Ascidian tadpole larvae looks very much like a chordate but there are main differences:
 - (a) Their larvae don't feed so some have no mouth.
 - (b) Segmentation is only in the myotomes of the tail.
- On hatching the larvae swim to surface like fish using the action of its myotomes against its notochord. It drifts a short distance in plankton, turns, swims down and searches for a suitable surface for attachment e.g. rocks. It then metamorphoses into an adult. It loses its notochord and tail but the gillslits expand tremendously.

Possible relationship of the ascidian larva to evolution of chordates.

Because an ascidian larva is the most primitive known chordate it appears that typical chordate features evolved not as adaptation in an adult but in a larva.

So in 1928 Walter Garstang, a British zoologist proposed that vertebrates evolved from a tadpole-like creature that failed to metamorphose. But became sexually mature when still a larva. This notion is now accepted by many zoologists and is supported by the facts that:

- A number of living animals develop some degree of sexual maturity while still larvae.
- Groups other than vertebrates also seem to have originated from young stages of their ancestors.



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11.3 Subphylum Cephalochordata

Members are commonly called AMPHIOXUS or LANCELETS. Only two genera with a total of 29 species are known to be still living. The genera

are:

- 1) Asymetron
- 2) Branchiostoma.

Whether they evolved from urochordates or from fishlike ancestors is debatable. However they look like simple fish with too many gill slits and look like tunicate tadpoles with the gill system.

Amphioxus

Two rings of tentacles around the entrance to the mouth screen out large particles and sense chemicals and touch. These tentacles edge the oral head. A velum surrounds the mouth opening. During feeding, cilia pull a current of water into the mouth and any food is filtered by a mucus net secreted by the endostyle which is a ventral groove in the pharynx. The endostyle also secretes iodine containing molecules and is homologous to the vertebrate thyroid. Food passes down to the gut where food is digested while water is pushed out through the gill slits. Gill slits don't lead directly to the exterior, but open into a large atrium, which has an opening to the exterior, called atriopore.

11.4 Subphylum Vertebrata

Vertebrates differ from other chordates in the possession of vertebral column, which replaces the notochord. Cephalization is pronounced with sense organs concentrated at the front end of the body, forming an obvious head. All vertebrates have:

- (1) Some sort of a liver.
- (2) Endocrine organs which secrete hormones.
- (3) Kidneys that differ completely from the various excretory organs of invertebrates.
- (4) A cranium (brain box) which supports and protects the brain and at least some traces of vertebrae.
- (5) The anterior end of the dorsal hollow nerve cord enlarges to form a brain. (6) A ventral chambered heart



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Primitively vertebrates were aquatic animals that moved as sharks of today (by sinusoidal swimming) with myotomes pulling against the notochord or vertebral column. Early fish fossils have no movable jaws and it is probable

the original feeding was by filtering as is seen in amphioxus. Although gills are used for gaseous exchange in modern fish, but it is Probable feeding was their original function in chordates. Vertebral column permitted efficient swimming and it is upon this that the success of fish depends.

Most vertebrate evolutionary advances have been made by carnivorous groups and involve improvements in locomotion and feeding. Radiation of terrestrial vertebrates was associated with evolution of the amniotic egg and the radiation of insects.

Subphylum vertebrata is made up of seven classes:

1. Agnatha
2. Chordrichthyes
3. Osteichthyes
4. Amphibia
5. Reptilia
6. Aves
7. Mammals

Effects of continental drift on the evolution of vertebrates

After the first vertebrates evolved continents drifted together and formed PANGAEA and then broke up into GONDWANNA and LAURASIA before forming the modern continents The movements and effects of them on climate greatly influenced vertebrate evolution e.g. much of Pangaea was near the equator a so early land vertebrate evolved in a warm climate. When Pangaea broke up landmasses moved north and south and Arctic Ocean cut off from warm ocean currents. The temperature over much of the land fell constituting to the extinction of dinosaurs. Later formation of C. America joining north and South America permitted migration that drastically affected vertebrate evolution in he Americans.

Use of CLADISTIC taxonomy is causing changes in vertebrate classification.

- Traditionally there are seven vertebrate classes, but cladistics argue that crocodiles are more closely related to birds than they are to turtles (or tortoises) or snakes.



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- For this reason reptiles should be divided into several classes. While this cladistics debate rages on till today (2003) we shall deal with the seven classes only.

11.4.1 Class Agnatha (Jawless fish): These are the hagfish and lampreys.

Earliest vertebrate fossils are the ostracoderms, fish without jaws from the Ordovician period (425-500myr) to the Devonian Period (345-405mya). Current hagfish and lampreys are the left relatives of these ostracoderms. They are scavengers and parasitic and found mainly in the warm Atlantic and Pacific waters.

Lampeys: They have a sucking mouth part and a rasping tongue covered with teeth. Unlike the hag fish they have a larval stage called the **ammocoete**. The adult lives only for a few years and dies soon after migrating upstream to sperm. Lampreys and hagfishes lack paired fins, jaws, scales and bones. The skeleton is just the notochord, which persists through out life, and various small cartilages.

Early Fish Evolution.

Ammocoete like amphioxus takes in food by filter feeding. But unlike in the amphioxus water current is moved by muscles which takes in more water and hence food. As a result an ammocoete can grow larger.

Without kidneys vertebrates could not live in fresh water where body tends to gain water by osmosis, or on land where the body tends to lose water by evaporation.

Lampreys and hagfish have kidneys and probably were the ostracoderms (which were marine); ostracoderms were pre adapted to move into fresh water lakes, which began to spread in SIRULIAN PERIOD. (405-425mya)

Chondrichthyes and actinopterygians (osteichthyes) became more successful during the Devonian period because:

1. Gill skeleton moved forward and evolved into jaws (biting and chewing)
2. Fins (pectoral, pelvic dorsal anal and caudal) were for balance and easy locomotion.
3. Jaws and locomotion permitted active lifestyle (catch and eat large prey).



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11.4.2 Class Chondrichthyes (Cartilaginous fish)

- They have a cartilaginous skeleton and occurred during the Paleozoic era (245- 570MYA) Cambrian, Ordovician, Silurian, Devonian, Carboniferous and Permian periods.
- Only 700 species survive today (Sharks, Dogfish, Rays, Skates, and Ratfish) which are all marine.
- They are ferocious, generally but the largest 10m long whale shark is filter feeder.
- Some rays have poison spines on their back or tail for defense.
- Shark's teeth are actually enlarged versions of its denticles found all over its skin.
- Sharks eat anything that comes their way and one killed in the Antarctic sea was found to have two rain coats parts of a horse cane number plate and rope. digestion is improved by an increase in surface area created by spiral valve which runs down the inside of the intestine.

Comparison of Cartilaginous and Bony Fish

Cartilaginous	Bony
Tail fin asymmetrical	Tail fin symmetrical
Pelvic fin placed far behind	Pelvic fin placed far forward
Have large fat filled liver used for buoyancy by alteration of lipid content.	Bony fish use gas-filled swim bladder for buoyancy.
Have denticles on the body.	Have scales on the body
Have internal fertilization (some ovoviviparous others are viviparous.)	Have external fertilization in most, only few show internal fertilization.
There is prenatal cannibalism in some e.g. in sand tiger (a shark)	

11.4.3 Class Osteichthyes (Bony fish)

Most fishes in this class have bone in their skulls, vertebrae, fin supports and scales. The gills are covered by a movable bony operculum on either side of the body. Usually, they



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have a lug of gas bladder and have paired fins and nostrils. The class is divided into two subclasses:

1. Subclass Actinopterygii (ray-finned bony fish):

- These are fish with skeleton of bones and have a versatile anatomy and physiology that bony fish gave rise to greatest diversity of vertebrates.
- Nearly all modern bony fish are ray-finned i.e. fins are supported by a ray of bones.
- Among the actinopterygians are filter feeders (e.g. herrings and parrotfish), insectivorous e.g. trout, carnivorous e.g. pike, and barracuda.

The success of bony fish is due to certain evolutionary innovations:

- Swim bladder: This is a gas filled sac formed as an outgrowth from the pharynx. By altering gas amount in the bladder the fish can maintain a desired depth in water. It is found in many actinopterygians but not in all. It is an accessory breathing organ and contributes to hearing, apart from taking part in sound production.
- Tail: Most of the push comes from the tail with paired fins being used for control.
- Gill slits: they don't open separately into the exterior as in sharks but instead they are covered by operculum. Oxygen laden water goes through the mouth, over the gills and out through the gill slits.
- They have spines on their dorsal fins or opercula for protection.
- They also have good sense organs, such as ears, lateral line organs, and eyes that are used in the aquatic environment. They have various body forms, adapted to their environment.

2. Subclass Sarcopterygii (lobe finned fish e.g. lung fish and coelacanth)

The pectoral and pelvic fins are fleshy. Terrestrial vertebrates evolved from this group except that it is not clear exactly from which genera this occurred. Only 5 lobe-finned genera survive today (4 genera of lungfishes occur in oxygen depleted freshwater swamps, and the coelacanth which is adapted to life in deep oceans). Lungfish have lungs, which supplement gill breathing the lungs and swim bladder both develop as outgrowths of the gut. Fossil records show that lungfish occurred during the Carboniferous period (286-360mya). They had fish-shaped bodies, short stubby legs and no gills.



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MOVE TO THE LAND

- Vertebrates evolved during the Cambrian period (over 500mya) when there was no plant life. During the mid Devonian when fish radiation was in

full swing, plants were spreading over swampy tropical Gondwana.

- These plants provided food for the vertebrates and by the end of the Devonian, insects were evolving and vertebrates had evolved from the Sarcopterygians (lobe finned fishes)

Several selective pressures contributed to the evolution of land vertebrate. These include:

- 1) Seas had many carnivorous fishes.
- 2) There was enough plant food on land and so vertebrates that were there had little competition.
- 3) Evolution of lungs.
- 4) Evolution of kidneys (produce concentrated urine.) and walls of digestive tract that absorb water help to reduce water loss from the body.
- 5) Evolution of sturdy bones and strong muscles.

Earliest fish fossil of a tetrapod is that of the late Devonian genus called *Ichthyostega* the earliest known Devonian amphibian. The morphological similarity between rhipidistian sarcopterygians and *Ichthyostega* suggests that amphibians evolved from the sarcopterygii:

- Primitive sarcopterygians possessed lungs and internal nares.
- Their fins were supported by bones corresponding to those of the tetrapod limbs
- The tail and ventral surfaces of *Ichthyostega* were covered with small scales similar to those of sarcopterygians.
- The caudal fin of *Ichthyostega* was supported by dermal fin rays as on sarcopterygians.
- The structure of the skeleton (vertebral column, cranium) of *Ichthyostega* was similar to that of rhipidistian sarcopterygians.
- The early amphibians were associated with freshwater deposits; rhipidistian sarcopterygians were freshwater carnivores.

11.4.4 Class Amphibia

Amphibians are tetrapods with smooth skins, devoid of hair or feathers. They depend primarily on external sources of heat in regulating their body temperature.



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General facts on Amphibians

- Amphibians have soft, moist and granular skin used for gas exchange despite the fact that most of them have small lungs
- Their aquatic larvae have gills.
- Only caecilians have scales
- The anterior end of the lateral line has developed into simple ears (a terrestrial life adaptation.) which respond to sound pressure.
- The first vertebrates to have evolved true tongues, which are long and sticky, and are shot out rapidly to catch flies.
- The third adaptation to life on land is the production of VASOPRESSIN (Anti diuretic hormone) which increases water absorption. But reproduction requires water i.e. eggs lack membranes and shell larvae. In some e.g. *pipa dorsalis* eggs develop in pouches on mothers back. Others e.g. *Rhinoderma* carry developing young in vocal pouch.
- Others like the female *Rheobatrachus* carry tadpoles in her stomach and the tadpoles release hormones that inhibit secretion of stomach acid that would digest them.
- Eggs of most frogs and toads hatch as miniature adults but some salamanders e.g. *Ambystoma* never undergoes metamorphosis i.e. paedogenesis or paedomorphosis. The eggs are unamniotic.
- Amphibians occur in large numbers only during mating but remain secretive most of the time, the male produces MATING CALLS and the female must recognize and approach the male of her own choice. Most hibernate on land or underwater in winter, but may also aestivate in summer.
- Most are protected by poison in the mucus in their skin in jelly surrounding their eggs. Colombian red Indians obtain a poison called batrachotoxin and by heating frogs of the genera *Dendrobates* and *Phylobates* and apply to tips of their darts and is less lethal to human.

There are three orders:

- 1) **Order Urodela:** There are about 350 species. They are more generalized and show transition from fish to tetrapod, they live in freshwater and moist environments. Their limbs contain small bones and muscles like those found at the base of lungfish fins.



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The hind legs are the same length as the fore legs. The ancestral tail is retained in adult life. They occur worldwide and include newts and salamanders.

- 2) **Order Anura:** There are about 3500 species and are tailless amphibians. They have specialized skeletons that give them ability to jump. The hind legs are more muscular than the fore legs and are used for swimming or jumping. They have a shortened backbone with many vertebrae that are fused together. The limb girdles firmly attached to the backbone, these species usually occur in freshwater, moist terrestrial habitats, there are some desert and brackish water species.
- 3) **Order Apoda (Gymnophiona):** There are about 170 species. These are elongated amphibians without legs i.e. APODA. They are mammal-like tropical animals adapted to burrowing in leaf litter or forestfloor. Fertilization is internal. They have reduced eyes and tails. They are also called Order Gymnophiona and the Caecilians.

11.4.5 Class Reptilia

The late Carboniferous (300mya) life took a different turn that saw rapid increase in tetrapod evolution, nests radiated widely onto land habitats, symbiotic evolution of insects and plants. About this time (300mya) enough animal life on land that supported predators explosive radiation of tetrapods (early tetrapods were probably carnivorous). Support for this is that all adult amphibians are carnivorous.

Early reptiles were less than 20cm long with stronger limbs and tucked further under the body to enable faster movement. Jaws were firmly attached to the skull enabling them to subdue and eat larger prey, waterproof skins covered with scales made of keratin and for this reason they breathed through the lungs. In tortoises and crocodiles scales are replaced as they wear away and in lizards and snakes scales are shed and replaced several times. The **amniotic egg** distinguishes later vertebrates from amphibians, and is suitable for terrestrial life.

Amniotic eggs

- Embryo is surrounded by membrane.
- Amnion encloses amniotic fluid



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- Two sacs (yolk and allantois) are attached to embryo
- Allantois stores wastes until hatching.
- Blood vessels grow from the embryo through the yolk sac and allantois to egg surface absorbs oxygen and release carbon dioxide.
- Chorion controls the overall permeability of the egg and chorion is in turn surrounded by the outer eggshell.
- Eggs are kept warm for rapid hatching and female pythons coil round the eggs for this purpose for three months; other snakes retain eggs in the body while others bury the eggs in sand or rotting vegetation.

Reptiles are tetrapods without feathers, hair or mammary glands. They have epidermal scales and few skin glands. Their development is direct and respiration is mainly by lungs. They are ectothermic, relying on external heat to regulate their body temperature. Usually, they have 5 digits, which end in claws. Reptiles have internal fertilization and the male has a penis, which may be branched. Most reptiles are oviparous, but ovoviviparous and viviparous species exist.

Classification of reptiles

Today's reptiles have four branches from the stem reptiles:

- Order Chelonia (Testudinata): This contains turtles and tortoises. They are herbivorous and are threatened by man's activity (hatching grounds), cannot be reared in quantity since sex is determined by eggs environmental temperature.
- Order Crocodilia: These are the closest relatives to dinosaurs. This order includes crocodiles in Africa, Asia and America. Alligators and Caimans of S.USA, China, Central America. Gavial of S.E Asia particularly in N. I India. They eat fish and have long snouts. They are all carnivorous and eggs are laid in nests, when newly hatched young ones occur. They spend much of their time in water and in water and nostrils permit them to breathe while the rest of the body is submerged..
- Order squamata: There are two suborders; lacertilia and Ophidia. The Suborder Lecertilia contains the lizards. Chameleon eats insects. The Suborder Ophidia has the snakes. They are legless and use muscles to throw the body into curves. They are swimmers, borrowers and climbers. Their ears don't open to the outside but respond to vibrations.



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- Order Rhynchocephalia: These are the tuatara lizards of New Zealand. They are currently endangered and are being protected.

11.4.6 Class Aves.

These are the only organisms with feathers. They occurred during the Jurassic period.

(150mya). They evolved during the Mesozoic era (66-245mya) as diaspids from the theropods (bipedal carnivorous dinosaurs with S-shaped necks). They differ from other theropods mainly in flight and endothermy.

Feathers are made from beta keratin, which is a protein related to the keratin of scales, hair, and horn.

- Feathers are flexible and provide excellent insulation.
- Colorful feathers are used in courtship displays and other types of social signaling.
- Feathers are moulted and replaced periodically as they become too damaged to function.

The hind limbs are free for running or swimming. Nearly all birds have two different types of locomotion (flight and swimming, running and flying, etc.)

Because birds must feed more often than reptiles both parents collaborate in nest building, egg incubation and feeding of the young. This led to extensive interaction between the sexes and has resulted in complex courtship, monogamy in some cases, or reproduction in large colonies. All birds lay eggs. The whole bird skeleton is light and strong for flight. Bones are mostly hollow. Horny bills and a gizzard replace heavy teeth. The lungs of bird connect to air sacs, which provide buoyancy and allow heat loss during flight.

Origin of birds

The earliest known bird is *Archaeopteryx* from fossils laid in the Jurassic period (150mya). *Archaeopteryx* is classified as bird because it had feathers or else it would have been classified a dinosaur. First Archaeopteryx was discovered in 1863 and raised a debate that still goes on i.e. the origin of the bird flight. There are three theories;

- Birds used broad-feathered forelimbs at first as stabilizers when jumping among branches and later as parachutes.



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- Birds were bipedal and the broadening and flattening of the fore limbs allowed them to make sustained jumps.
- Feathers evolved as insect's nets used to scoop flying prey to the mouth.

Archaeopteryx is believed to have originated from reptiles. It shares a number of features with reptiles:

- Possession of teeth
- Wings had clawed fingers
- Possession of a muscular reptilian type of tail
- It had scales on the feet. Reptilian bodies are covered with scales
- Feathers of birds are believed to have originated from reptilian scales.

The class is divided into two subclasses: Archaeornithes (ancient birds) and Neornithes (recent birds).

11.4.7 Class Mammalia.

Mammal's features evolved in various lines of synapsid reptiles during the Permian (250mya) but mammals and dinosaurs both appeared during the Triassic period (about 240mya). By late Mesozoic (about 200mya), monotremes, marsupials and placental mammals had evolved.

A mammal can be defined as vertebrate with only one bone on either side in the lower jaw. This feature allowed a carnivore to grip a struggling victim more firmly. Other features are possession of hair, important in thermoregulation and mammary glands which are for milk production to nourish the new born. They are endotherms, using heat from their metabolic activities to regulate body temperature. Mammals give birth to live young (viviparity).

Specialized teeth evolved (molar, canines, and incisors). The incisors are used for cutting, canines for gripping and tearing and molars for crushing and breaking. Mammals are therefore adapted to herbivorous, carnivorous, omnivorous and insectivorous diets among others. Early mammals were small the size of modern mouse or shrew. Endothermy perhaps allowed them to avoid competition with reptiles by being active at night when it was cold. Viviparity enabled them to remain mobile while incubating embryos. Mammalian integument has many roles, and consists of temperature control, protection, excretion and communication. Adaptations for fast quadrupedal locomotion include a narrow foot track, mobile limb girdles, lengthening of the stride, and clustering of muscles at the top of the leg. The tail is freed from attachment to the hind leg and is used for other purposes like balance, as a fifth hand or as a fly swatter. Success



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of mammals led to extensive adaptive radiation during which herbivours bipedalism, flight and many other adaptations evolved.

Classification of mammals

There are three main groups of mammals: the monotremes, Marsupials and the Placentals.

The placentals have 19 orders, together with the monotremes and marsupials give 21 orders.

Subclass Prototheria

Order: Monotremata: Only three species are living i.e. the dark-billed platypus, and two kinds of echidnas (spiny anteaters). They lay eggs, but females feed the hatched young ones on milk. They are toothless as adults. Monotremes provide evidence that viviparity and milk production didn't evolve at the same time. Eggs laid on the ground are kept warm by the body. Australian platypus lives as water rat at the edge of the streams and swims using its flattened tail.

Subclass: Theria

Infra class: Metatheria: Members do not have a true placenta

Order: Marsupialia; The young are born at an early age and finish development in a pouch or marsupium. They are found only in Australia and America (e.g. the Opossum.). Australian marsupials include Kangaroos and Koala. Birth in kangaroos takes place eight days after fertilization and the young crawl into the marsupium where the young attach to a teat and muscles around the mothers mammary gland pump milk into them while they develop.

Infra class: Eutheria (True placental mammals):

They have a true placenta. Their eggs have no yolk at all. Most modern mammals e.g. goats are placental. The mother's blood supplies food and oxygen. Many have developed social systems, which probably evolved when both parents cared for and fed the young. They are classified into 19 orders and are adapted to many ways of life following adaptations by earlier environments. Arranged from the earliest evolved Rodentia, Lagomorpha, Insectivore Primate, Chiroptera, Carnivora, Pinnipedia, (senes), Proboscidea, Perissodactyla, Artiodactyla and Cetacea.

Revision questions

I. Describe the five-echinoderm classes.



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- II. Explain how the water vascular system operates in sea stars
- III. What are the functions of ciliated tube feet of sea stars?
- IV. Distinguish between sea stars and brittle stars. sand dollar and Aristotle's lantern.
- V. What feature led to the miscalculation of phylum hemichordata (Acornworm) as a chordate?
- VI. Outline the five major chordate features.
- VII. What feature makes us classify Urochordates as Chordates and what plant feature does he possess.
- VIII. Explain Garstang hypothesis on the evolution of chordates from tadpole-like creature.
- IX. Discuss subphylum Cephalochordata.
- X. Describe the effects of continental drift on the evolution of vertebrates.
- XI. Outline the early fish evolution and give the main differences between Chondrichthyes and Osteichthyes. Actinopterygians and Sarcopterygians.
- XII. Explain the selective pressures that contributed to the evolution of land vertebrates.
- XIII. Outline; classification of amphibians, and the natural history of amphibians.
- XIV. Describe amniotic eggs, and the classification of reptilia.
- XV. Distinguish between: lacertilia and ophidia, chelonia and Rynchocephalia, feathers and Patagium, Cetacea and Pinnipedia, Chiroptera and Artiodactyla, Perissodactyla and primates.
- XVI. Explain why man has dominated all other animals despite the fact that he is not as advanced evolutionarily as, say, goats.



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LESSON 12 EXCRETION

The term excretion means separation and elimination of the unwanted waste materials from the body. The waste materials are of several kinds and vary from animal to animal and also in the same animal from time to time.

Animals are broadly classified into three groups, based upon the type of nitrogen excretion.

- 1) **Ammonotelic animals.** Animals are described as being *Ammonotelic* when nitrogen is excreted predominantly in the form of ammonia. This is especially true in most of the aquatic animals i.e. certain protozoan, actinozoans, polychaete annelids, crustaceans, *Aplysia*, *Sepia*, *Octopus*, among molluscs, teleosts, tadpoles of amphibia and crocodiles
- 2) **Ureotelic animals:** When nitrogen is excreted predominantly in the form of urea, the subjects are said to be *ureotelic* e.g. amphibians and mammals. In fishes both ammonotelic and ureotelic metabolisms are met with.
- 3) **Urecotelic animals.** Urecotelic animals are those in which nitrogen is excreted predominantly in the form of uric acid e.g. insects, some gastropods, lizards, snakes, lizards and birds.

Excretory organs in animals

Among animals special organs are usually present in the body for the elimination of nitrogenous waste, called the excretory organs. In the primitive lowly organized members of the animal kingdom, excretion is through the body surface by simple process of diffusion. Among the higher animals, there is the excretory organ that performs two functions i.e. Elimination of waste material, and Maintenance of osmotic pressure.

Excretory organs in invertebrates.

Protozoa: Waste products are removed by the contractile vacuole. This organ is looked upon as an osmoregulatory organelle. Its function is to remove the excess water that has entered into the protoplasm by endomosis. Excretion is by simple diffusion.



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Porifera and coelentrata: Excretion is effected by direct diffusion from the body into the external medium.

Platyhelminthesis: Have an excretory system called the flame cell.

Nematoda: Excretory system is in the form of a longitudinal tube with or without the flame cells.

Annelida: Nephridia are the excretory organs of annelids. They may be:

- a) Protonephridium or
- b) Metanephridium.

Arthropoda: This depends on the class e.g. melacostraca use the antennary glands, entecostraca use maxillary glands, insects use malphigian tubules.

Mollusca: The Nephridia are usually thought to be he organs of excretion.

Echinodermata: Specialized excretory organs are absent. Excretion is effected by the amoebocytes found in the coelomic fluid.

The principle source of nitrogen is protein of the food and in all the animals the liberation of ammonia from the proteins takes place as follows:

Proteolytic enzyme deamination(liver)

Protein_____aminoacids_____ammonia

Animals are called as Ammonotelic ureotelic and uricotelic, depending upon the type of nitrogenous excretory materials excreted.

Ammonotelic Animals

Animals are described as Ammonotelic when nitrogen is excreted predominantly in the form of ammonia. It is especially true in most of aquatic animals ie protozoans, and most molluscs and teleosts. Aquatic invertebrates excrete nitrogen in form of ammonia because the solubility of ammonia is very high and it can readily diffuse out into the surrounding water medium through the body surface. Several experiments have shown that:

- If crystalline urease is injected into reptiles, ammonia is formed by hydrolyzing small amounts of urea present in the blood and if concentration rises above 20,000 death results. Thus ammonia requires large amounts of water for its excretion.



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- *Spirostomum* produces a large quantity of ammonia when the oxygen tension is less than that of the air
- Various Echinoderms e.g. starfish, sea urchins and sea cucumbers are Ammonotelic but they also give considerable amount of aminoacids in to the water.
- Among arthropods, ammonia seems to predominate as an excretory product in the fresh water as well as marine crustaceans.
- Among molluscs, the cephalopods, *Sepia* and *Octopus* excrete most of their nitrogen as ammonia.
- Earthworms are intermediates between aquatic and terrestrial. The following experiments can illustrate this: earthworms kept under moist air produce more urea than ammonia. If they are immersed in water they become Ammonotelic.
- Most of the freshwater fishes are Ammonotelic though they excrete some urea also. Most of their nitrogenous excreta diffuses out through the gill surface.
- Nitrogenous excretion in lungfishes have been studied e.g. the *Protopterus* when these fishes are active in water they excrete nearly more nitrogen as ammonia than as urea. When aestivated in mud cocoon, however it excretes urea. When the fish returns to water urea is rapidly excreted and excretion of nitrogen as ammonia increases to its characteristic high proportion.
- In most Ammonotelic animals, the ammonia is flushed out of the body by water either along the body surface or through the gills.

Trimethylamine Oxide is the nitrogenous compound occurring in the tissues and urine of marine teleosts. This is responsible for the unpleasant smell of dead marine fish.

Ureotelic animals

By the retention of nitrogenous waste in the form of urea, the ureotelic animals should have raised their blood concentration to same levels as that of the surrounding seawater. In the course of ages, elasmobranchs have changed their metabolism to the production of more urea, so that today all tissues except the brain and blood are able to synthesize urea, and sharks have 2 to 2.5% urea in their blood and excrete nearly 80% of their nitrogenous waste as urea.

Excess of urea is excreted through the kidneys. The conversion of such a large amount of



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nitrogen into urea reduces the percentage of ammonia in urine to 8%. The sharks must, therefore be ureotelic. Their tissues are so loaded with urea that their condition is ureamic. The urea is maintained by the fact that the

gills are impermeable to urea and a considerable part of the urea in the urine is reabsorbed into the blood by renal capsules.

Amphibia

The livers of both frogs and toads can synthesize urea and the cells of their renal tubules appear capable of excreting urea. The tadpole is Ammonotelic like the fishes, over 40% of their nitrogenous excretion being in the form of ammonia. As the tadpole metamorphoses into an adult, ammonia decreases to 12.5 % and urea increase proportionately. This is an adaptation to life on land, where the adult frog cannot have as much water as the tadpoles get. Urea is much diffusive and soluble substance like ammonia, but differs from it, in that it is less toxic even in relatively high concentration and hence can be retained in the body for some time before being eliminated.

This feature is advantageous to amphibian's animals. Production of urea is an adaptation to condition of water shortage, the amphibia had to face on land, just as the elasmobranchs had to face in the marine habitat

Mammals

Even the most primitive mammals i.e. the monotremes are ureotelic. In these truly terrestrial animals, metabolism is ureotelic from the embryonic period onwards. Being viviparous, they are in connection with the maternal blood stream throughout the embryonic life. The water supply is therefore unrestricted as in the aquatic habitat. Ammonia resulting from the nitrogen metabolism is detoxicated through the formation of urea, which diffuses into maternal kidneys. In the adult, however, closely correlated with the control of the body temperature through the production of sweat and water regulation in general, the body maintains enough water to flush out urea.

Uricotelic animals

Uricotelic animals are those that excrete nitrogenous waste mainly in the form of uric acid. Among the invertebrates, the insects are ureotelic. These groups of animals are perfectly



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terrestrial but, unlike the mammals, they lack not only an efficient system of water regulation but also viviparity. Their early development takes place within the narrow confines of cleodolic (closed box-like) egg, within

which the nitrogenous waste has to be collected in a concentrated form, which does not require much water for its removal. Examples are insects, few reptiles and birds as whole.

Reptiles: among the modern, reptiles, the lizards and snakes live in dry areas, where water conservation is important. Further, their eggs must retain the embryonic nitrogenous wastes till they are hatched. Therefore, their nitrogenous excretion is in the form of relatively insoluble, nontoxic acid, which in house lizards and tree snakes, is in the form of solid pellets.

Birds: birds as a whole group, are uniform in their physiology. This is very true with their nitrogen excretion. There is about 10% of urea and 3.4% of ammonia in their urine, indicating the fact that in no animal nitrogenous excretion is in the form of a single product, and that ammonia, urea or amino acid are likely to break through the complex chemical processes of nitrogen metabolism. However the birds are mainly ureotelic and their fecal matter also includes the solid uric acid.

Insects: the bug *Rhodnius* excretes solid urine consisting of 90-92% uric acid in crystalline form with no ammonia and no aminoacids, and only a trace of urea. This is true of other few insects to whom supply is too restricted. In several forms the different products vary in amount. *Rhodinus* secretes more urea immediately after a meal of blood, and the blowfly gives off more ammonia after eating meat. However it is difficult to be sure if these products are from the gut or are true metabolic wastes, in as much as the malphigian tubules open to the proctodaeum.

Gastropods: snails though Ammonotelic, have amounts of uric acid varying with the water supply of their environment. Limnaea lives in water, but lays cleidoic eggs on land and the embryos excrete 4.4% mg of uric acid. During hibernation the nephridium stores as much uric acid as 0.75 of the weight of the organ. Terrestrial snails on the other hand, have larger



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amounts of uric acid in the kidneys of the adults, and those that have secondarily taken to water have relatively more uric acid than the strictly marine forms.

Other Invertebrates: even in Ammonotelic forms, uric acid has been found to occur in the blood of coelomic fluid or in the tissues. In chaetopterus it amounts to 0.73mg in 100ml and in the crayfish 2.3mg in 100ml of the urine. Since even in the mammals (which are ureotelic) as much as uric acid as 0.23mg per ml of urine may occur, the distinction between these and the truly ureotelic animals, like birds, appear to be based on quantitative differences in the amount of uric acid produced through the nitrogen metabolism.

Revision questions

- I. Give a classification of animals based upon the type of nitrogen excretion.
- II. Outline the excretory organs found in the invertebrate phyla.
- III. Distinguish between: (a) ammonia and trimethylamine oxide.

(b) ureotelic and Ureotelic animals.



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LESSON 13 OSMOREGULATION

Hober in 1902 first coined the term osmoregulation to the collective activity of the variety of mechanisms used by organisms to control water movement and water volumes. This implies the maintenance of an internal osmotic concentration which is different from the surrounding water medium.

The internal ionic concentration is also to be regulated as large differences exist in the external ionic concentration as well as the ionic concentration in the different portions of the body of the animals. The various processes which he maintains internal ionic concentration of the body fall under **ionic regulation**. The two processes are inseparable and sometimes the term osmoregulation is used to refer to the two processes.

Osmoregulation in different animals

- **Protozoa:** protozoa's either fresh water or marine don't face any osmotic problem. Marine and parasitic protozoa are isosmotic with the medium, and eliminate only water which enters along with the food material. Fresh water protozoan are hyperosmotic to their medium and the contractile vacuole eliminates excessive water. Some of the protozoan tolerate extremities in the medium. *Amoebae lacerata* lives freely either in freshwater or seawater.

Volume Regulation.

Osmotic regulation in protozoan is effected by volume change and vacuolar output. The volume of *Amoebae proteus* and *polymxa* has been measured by putting them in capillaries. *Amoeba mira* and *Amoeba lacerata* which are marine and brackish water forms, show good volume regulation within a few hours when transferred to fresh water. When the concentration of the external medium is changed, the fresh water forms, fail to return to their normal shape for at least two hours.

Entrance of water.



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The surface of the protozoan is permeable to water. As a result of the entry of water, the animals swell. This permeability is measured in volume of water entering per unit of the cell membrane for an osmotic gradient in a given period of time. Water may enter inside the protozoan not only from the surface, but also during feeding. In peritrichous ciliates, inhabiting fresh water, the water taken up by the food vacuoles may be 8 to 20% of the output of the contractile vacuole and in paramecium 30%. In paramecium the activity of the contractile vacuole increases by merely pumping.

Function Of the Contractile Vacuole

In freshwater inhabitants belonging to the four groups of protozoa, a contractile vacuole is invariably present, while it may be present or absent in marine and endoparasitic forms. Sporozoans don't possess a contractile vacuole. In the marine and parasitic protozoan, the contractile vacuole eliminates metabolic water and water that enters along with the food material. Besides this, some water is required to dispose off the excess of salts.

The output of water from the contractile vacuole in freshwater protozoan is more than the marine or endoparasitic forms. The period of time required to eliminate a quantity of water equal to the body volume of the animal for about 15 freshwater forms ranged between 4.1 to 53 minutes.

Fresh Water Crabs:

Fresh water crabs live in medium that is hypotonic to blood, i.e. the salinity of the blood is more than that of the seawater. The adaptations listed for animals living in hypotonic medium occur in these animals. They are;

- i. More water is excreted in the form of dilute urine. However the urine, though hypotonic to blood, is hypertonic to fresh water. Thus some of the salts are lost.
- ii. The loss of salts is made good by kidney or antennary gland, where reabsorption of salts takes place.
- iii. An ionic regulation takes place when the blood is concentrated, more potassium and calcium and less magnesium are present than in the medium suggesting a regulation of mineral composition of blood.



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iv. Salts are also procured from food material.

v. Absorption of salts takes place by gills also.

Through all these devices the salt concentration of the internal medium is maintained.

• Osmoregulation in brackish water animals.

Brackish waters are mixohaline ranging between 30 and 0.5 salinity. These waters are restricted to coastal regions, such as estuaries or salt marshes, where the salinity often changes with the tides. The brackish water forms can be divided into three groups,

- i. Marine forms which tolerate low salinity
- ii. Fresh water animals which can tolerate moderate salinity
- iii. True brackish water animals, which are not found either in sea or fresh waters. In brackish water forms like *Carcinus* which osmoregulates, the blood concentration corresponds to about 60% seawater when it is living only in 15% seawater. When the concentration of body fluids is higher than the surrounding medium, water enters by osmosis and salts are lost by diffusion. Therefore to maintain balance, water must be quickly removed from the body and solutes must be replaced. Water is removed by urine and through other sources.

• Estuarine Crabs

Estuarine crabs such as *Carcinus* are euryhaline and homeiosmotic. As the water from rivers enters the sea in the estuaries, the medium becomes dilute and wide fluctuation occurs. If *Carcinus* is transferred to 75% seawater there is only a 0.3% increase in weight and then it returns back to normal within 12 hours.

Hypo-Osmotic Regulators

Some animals maintain the salt concentrations of their body fluids below that of the surrounding environment. Such animals occur both in seawater as well as in inland waters of high salinity. Animals which live in inland saline waters can be divided into two groups:

- i. Animals found typically in brackish water or moderate salinity, low salinity or even in fresh water.
- ii. Animals which exist in high salinity crustaceans like *Palaemonetes* a gastropod and the fish *Cyprinodon fasciatus* are examples of the first group while the brine shrimp, *artemia salina* belong to the second group.



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The composition of the body fluids of animals which follow hyposmotic regulation varies with the concentration of the medium. In marine and salt-water vertebrates, the blood concentration is only slightly higher than that

of fresh water forms.

The osmotic pressure of blood is due to sodium and chloride ions whose concentration is higher.

Salt and water balance.

The main problems in hyperosmotic medium are

- I. Loss of water by exosmosis,
- II. Gain of salts by diffusion

To maintain equilibrium the animals should excrete salts and conserve water. Salt is removed by secretion into surrounding water. Teleosts and brine shrimp secrete salts into the surrounding medium without any loss of water.

The water lost is replaced by food or by drinking the water in the surroundings. Salt gained is excreted and water is preserved.

Animals that live in hypertonic environment include: marine teleost fishes, marine crabs, a number of prawns, larvae of dipteran insects and the brine shrimps which live in salt lakes.

Salt excretion in marine teleosts.

The teleost fishes living in seawater face the problem of loss of water. Their blood is hypotonic to seawater. The skin of these fishes, as it is covered over by scales and a coat of mucus, is impermeable and water escapes through the gut and gills.

The water lost osmotically is recovered by drinking seawater and excreting the salts extra-renally. Kidneys can not excrete urine that is hypertonic to the blood, thus salts leave the body through urine, more than that enters the body. The extra salts are excreted from the body by chloride secretory cells situated in the gills.

Osmoregulation in marine crabs.

Marine crabs whose blood concentration varies with the fluctuations in seawater are stenohaline e.g. the *Maia* and *Palinurus*. In these animals the renal organs are well



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developed and occur on the first segment of their second antenna. They are called the green glands or the antennary glands. They excrete urine in large amounts and this helps remove chlorides, sulfates calcium and magnesium.

Osmoregulation in migratory fishes.

In migratory fishes like *Anguilla* both the adaptations to live in seawater and fresh water occur. These fishes that live in fresh water lakes migrate to the sea through rivers and in the sea they may travel upto 1,000 miles. The young larvae of these fishes ,the elvers, again migrate to he fresh water lakes. The eel have to regulate their concentration of blood both in fresh water and in seas. In marine it is maintained by drinking seawater and secreting the extra salt through the chloride cells and making the urine isotonic with he blood.

In fresh water the concentration which is less than in the sea is maintained as in fresh water fishes. Water in large amounts is removed in urine and the salts are reabsorbed.

Osmoregulation in terrestrial animals

The osmo and ionic regulation in terrestrial animals is similar to that of marine animals. They are surrounded by a medium which contains a low concentration of water than the body fluids and obtain salts readily through food material. Both these animals have to conserve water and excrete salts.

Conservation of water.

Terrestrial animals have a body covering that is impermeable to water except those living in places where the atmospheric humidity is high. The respiratory surfaces, are enclosed within body cavities so that water vapor is conserved.

In warm-blooded animals the sweat glands secrete water, which withdraw the body heat during its evaporation. the area that can secrete sweat are limited and are controlled by the nervous system.



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Reabsorption of Water

The tubules in the kidneys of terrestrial vertebrates have the capacity of reabsorbing water from the urine .almost 99% of the water in the urine is reabsorbed in he tubules of mammals.

Osmoregulation in amphibia.

In amphibians water enters osmotically at a slow rate due to the low permeability of skin. The water which enters is excreted as dilute urine. Salt is lost in he urine but is again absorbed from he pond water. The loss of salts is also minimized by active absorption in the kidney tubules and urinary bladder. Passive water reabsorption also occurs which increases the concentration of urine of some waste products that are not absorbed. The tubules of the kidney may also excrete some specific salts such as urea. Urine production is under control of the neurohypophysis, which may altogether stop urine production when the animal stays on land

Osmoregulation in reptiles and birds

Scales cover the reptilian skin and loss of water is thus minimized. The water content of desert animal is maintained by the conservation of water. The volume of urine excreted is small in terrestrial forms which may even be semisolid. The cloaca may reabsorb water. In land reptiles loss of water is more by way of lungs and desert reptiles get most of their water from food. Lizards and snakes often take water to compensate its ;loss. Marine reptiles have special glands in the head with openings in lachrymal of nasal ducts, these glands secrete salts and thus the excess salts of the body are removed.

In birds the cloacal urine is viscous paste of uric acid crystals. Waste absorption may take place in the cloaca, lower intestine and kidney.

Osmoregulation in mammals.

Regulation of water content, permits mammals to live in moist or dry air, in fresh or salt water and over a wide range of temperatures. The kidney tubules produce a blood hyperosmotic urine. Endocrine and nervous systems regulate osmoregulation in mammals. Urine production in man is dependent on the temperature and body water load.



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The permeability of the skin to water is extremely low. Many mammals however use water evaporation by sweat glands as a means of cooling.

Water reabsorption also takes place in the kidneys. Those mammals which produce the most concentrated urine have the longest loop of henle as absorption takes place in this part. In man, 1,100 liters of plasma circulated per day in the glomerulus yields 180 liters of filtrate, out of which 178.5 liters are absorbed leaving 1.5 liters per day of urine.

Revision questions.

1. Describe briefly the following:

- (a) Osmoregulation in protozoans.
- (b) Osmoregulation in fresh water crabs.
- (c) Osmoregulation in brackish water animals.
- (d) Osmoregulation in estuarine crabs.

2. Write brief notes on:

- (a) hypo-osmotic regulators.
- (b) Osmoregulation in: (i) Marine teleosts (ii) marine crabs.
- (iii) migratory fishes
- (iv) amphibians